

Wannier90 / WannierBerri를 사용하여 Wannier 내삽 방법을 이용한 물성 계산

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Bloch & Wannier functions

i, j Slowly varying in $\mathbf{k}$
Enables interpolation in $\mathbf{k}$

m, n Energy eigensates

$$|u_{i\mathbf{k}}^{(W)}\rangle = \sum_m |u_{m\mathbf{k}}\rangle U_{mi\mathbf{k}}$$

Square, unitary matrix

$$|u_{m\mathbf{k}}\rangle = \sum_i |u_{i\mathbf{k}}^{(W)}\rangle U_{im\mathbf{k}}^\dagger$$

Periodic part of Bloch sum $\langle \mathbf{r} | u_{i\mathbf{k}}^{(W)} \rangle = \frac{1}{\sqrt{N_{\mathbf{k}}}} \sum_{\mathbf{R}} e^{-i\mathbf{k} \cdot (\mathbf{r} - \mathbf{R})} w_i(\mathbf{r} - \mathbf{R}) : \quad \mathbf{r} \rightarrow \mathbf{r} + \mathbf{R}'$

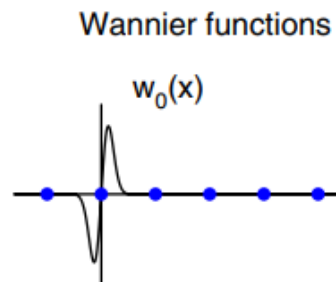
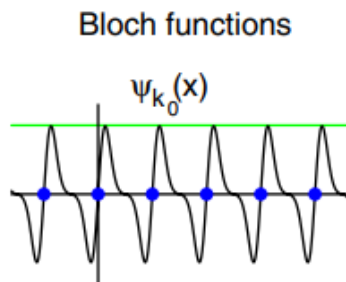
Normalization convention (Integration over supercell)

$$|u_{i\mathbf{k}}^{(W)}\rangle = \frac{1}{\sqrt{N_{\mathbf{k}}}} \sum_{\mathbf{R}} e^{-i\mathbf{k} \cdot (\hat{\mathbf{r}} - \mathbf{R})} |w_{i\mathbf{R}}\rangle \approx \frac{1}{\sqrt{N_{\mathbf{k}}}} e^{-i\mathbf{k} \cdot \mathbf{r}_i} \sum_{\mathbf{R}} |w_{i\mathbf{R}}\rangle$$

Localized in $\mathbf{r}$

$$|w_{i\mathbf{R}}\rangle = \frac{1}{\sqrt{N_{\mathbf{k}}}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot (\hat{\mathbf{r}} - \mathbf{R})} |u_{i\mathbf{k}}^{(W)}\rangle$$

If $w_{i\mathbf{R}}$ localized at $\mathbf{r} = \mathbf{R} + \mathbf{r}_i$



Projection: An easy way to find smooth functions

$$|\phi_{i\mathbf{k}}\rangle = \sum_{n=1}^{\tilde{N}_{\mathbf{k}}} |\psi_{n\mathbf{k}}\rangle \langle \psi_{n\mathbf{k}} | g_i \rangle = \sum_{n=1}^{\tilde{N}_{\mathbf{k}}} |\psi_{n\mathbf{k}}\rangle A_{ni\mathbf{k}}$$

$N \leq \tilde{N}_{\mathbf{k}}$ initial guesses

Insensitive to any unitary rotation

→ Smooth in $\mathbf{k}$

$$A_{ni\mathbf{k}} \equiv \langle \psi_{n\mathbf{k}} | g_i \rangle$$

$$S_{ij\mathbf{k}} = \langle \phi_{i\mathbf{k}} | \phi_{j\mathbf{k}} \rangle = (A_{\mathbf{k}}^\dagger A_{\mathbf{k}})_{ij}$$

$$|\tilde{\phi}_{i\mathbf{k}}\rangle = \sum_{j=1}^N |\phi_{j\mathbf{k}}\rangle S_{ji\mathbf{k}}^{-1/2}$$

Lowdin orthonormalization: Better than Gram-Schmidt

If $\tilde{N}_{\mathbf{k}} = N$, just a unitary rotation.

If $\tilde{N}_{\mathbf{k}} > N$, then choosing N states out of $\tilde{N}_{\mathbf{k}}$ states.

$$\langle \tilde{\phi}_{i\mathbf{k}} | \tilde{\phi}_{j\mathbf{k}} \rangle = (S_{\mathbf{k}}^{-1/2} S_{\mathbf{k}} S_{\mathbf{k}}^{-1/2})_{ij} = \delta_{ij}$$

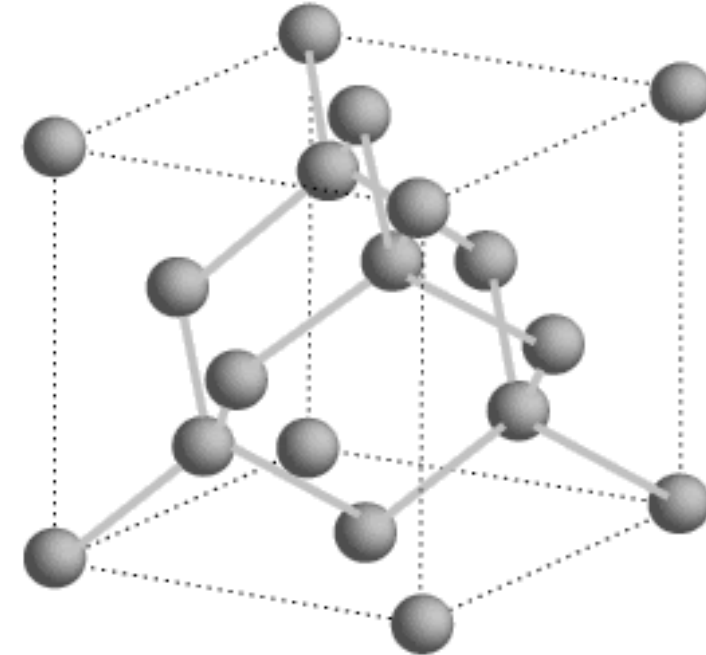
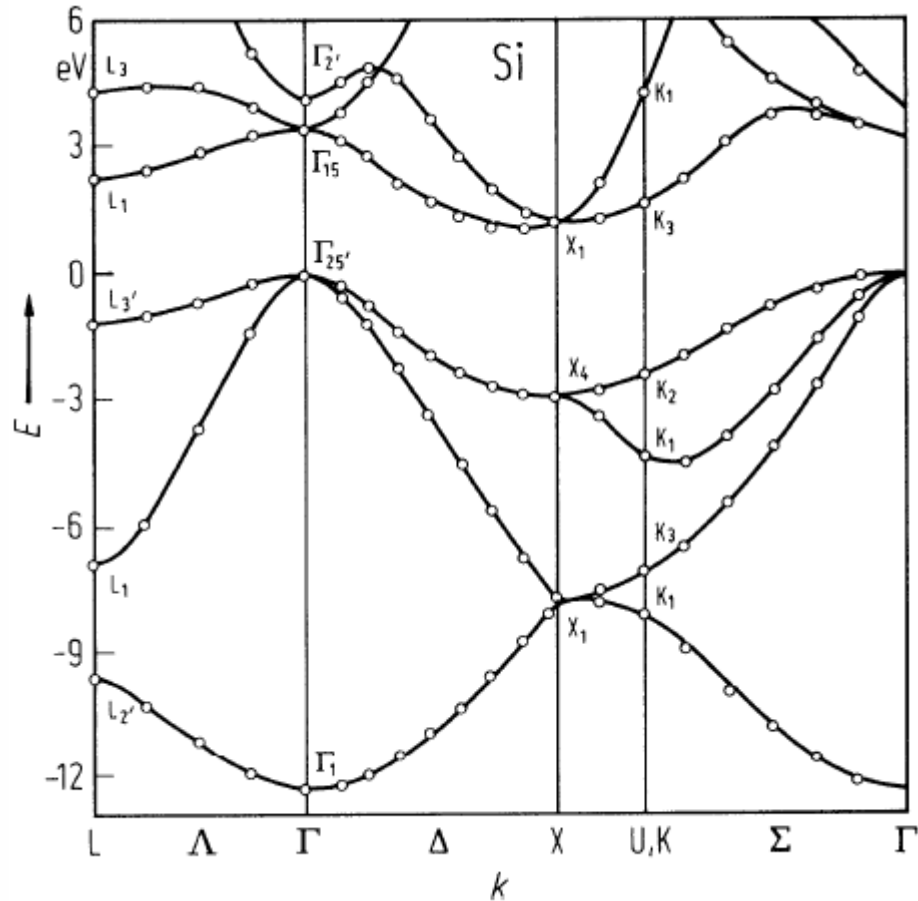
Projection: Bonding, anti-bonding eigenstates

$$|\psi_{\pm\mathbf{k}}\rangle = \frac{1}{\sqrt{N_{\mathbf{k}}}} \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} \frac{|g_{\mathbf{L}\mathbf{R}}\rangle \pm |g_{\mathbf{R}\mathbf{R}}\rangle}{\sqrt{2}} e^{i\theta_{\pm\mathbf{k}}} \quad \text{Not smooth in } \mathbf{k}$$

$$|\phi_{\mathbf{L}/\mathbf{R}\mathbf{k}}\rangle = \sum_{n=\pm} |\psi_{n\mathbf{k}}\rangle \langle \psi_{n\mathbf{k}} | g_{\mathbf{L}/\mathbf{R}0} \rangle = \frac{1}{\sqrt{N_{\mathbf{k}}}} \frac{|\psi_{+\mathbf{k}}\rangle e^{-i\theta_{+\mathbf{k}}} \pm |\psi_{-\mathbf{k}}\rangle e^{-i\theta_{-\mathbf{k}}}}{\sqrt{2}} = \frac{1}{N_{\mathbf{k}}} \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} |g_{\mathbf{L}/\mathbf{R}\mathbf{R}}\rangle$$

Smooth in $\mathbf{k}$
Similar character as the projector

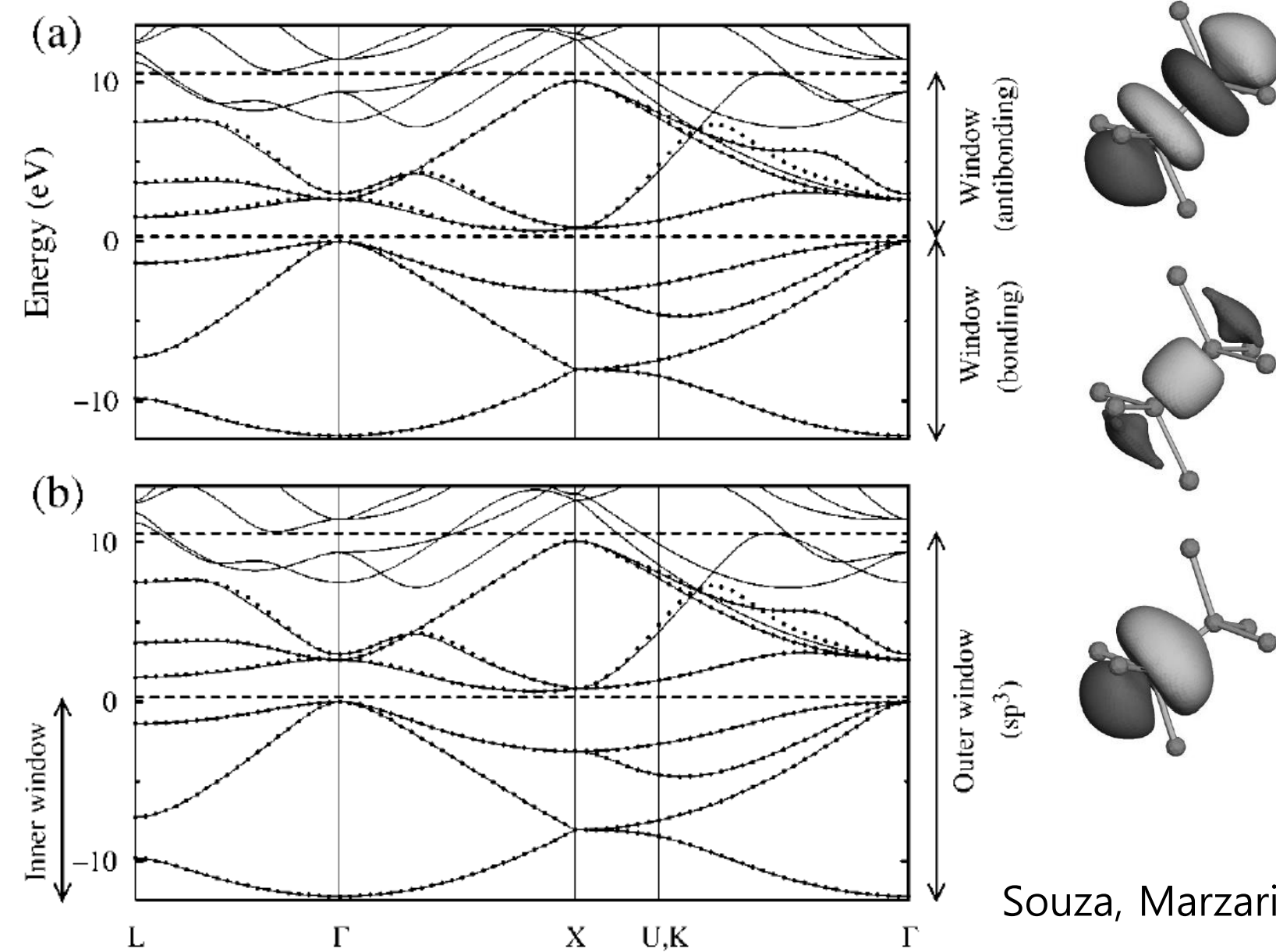
Silicon



https://lampz.tugraz.at/~hadley/ss1/semiconductors/silicon_bandstructure.php

<http://hyperphysics.phy-astr.gsu.edu/hbase/Solids/sili2.html>

Silicon: Projection

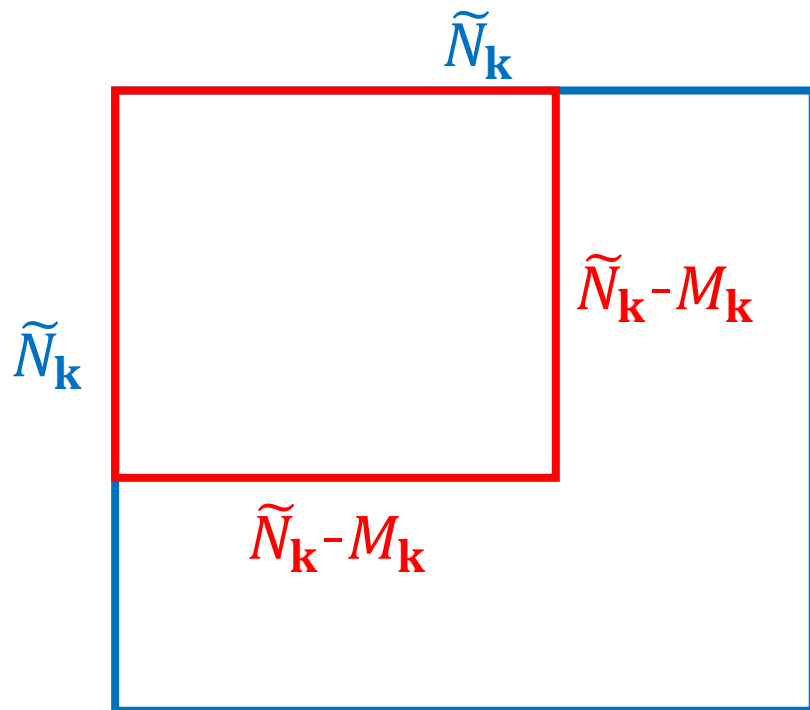


Souza, Marzari, Vanderbilt, PRB **65**, 035109 (2001)

Projection + inner/outer window

$$\tilde{P}_{\mathbf{k}} \equiv \sum_{i=1}^N |\phi_{i\mathbf{k}}\rangle \langle \phi_{i\mathbf{k}}|$$

$$\tilde{P}_{mn\mathbf{k}} = \sum_{i=1}^N \langle \psi_{m\mathbf{k}} | \phi_{i\mathbf{k}} \rangle \langle \phi_{i\mathbf{k}} | \psi_{n\mathbf{k}} \rangle$$



Eigenvalues: N 1's and $(\tilde{N}_{\mathbf{k}} - N)$ 0's

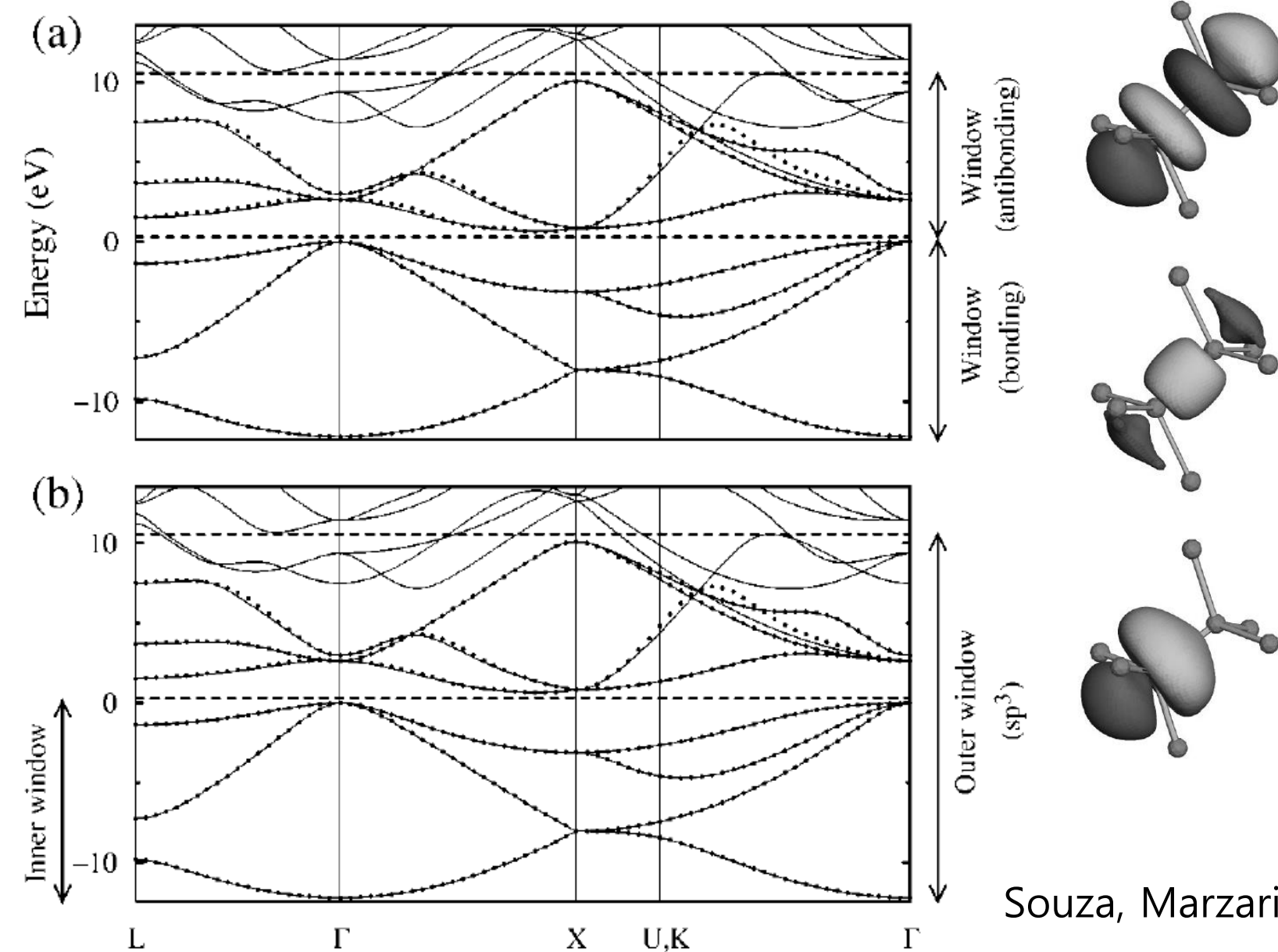
Choose $M_{\mathbf{k}}$ states in the inner window

Diagonalize the smaller matrix:
eigenvalues are between 0 and 1

Find $N - M_{\mathbf{k}}$ eigenvectors with largest eigenvalues

Do the projection with those N states
→ Smooth functions

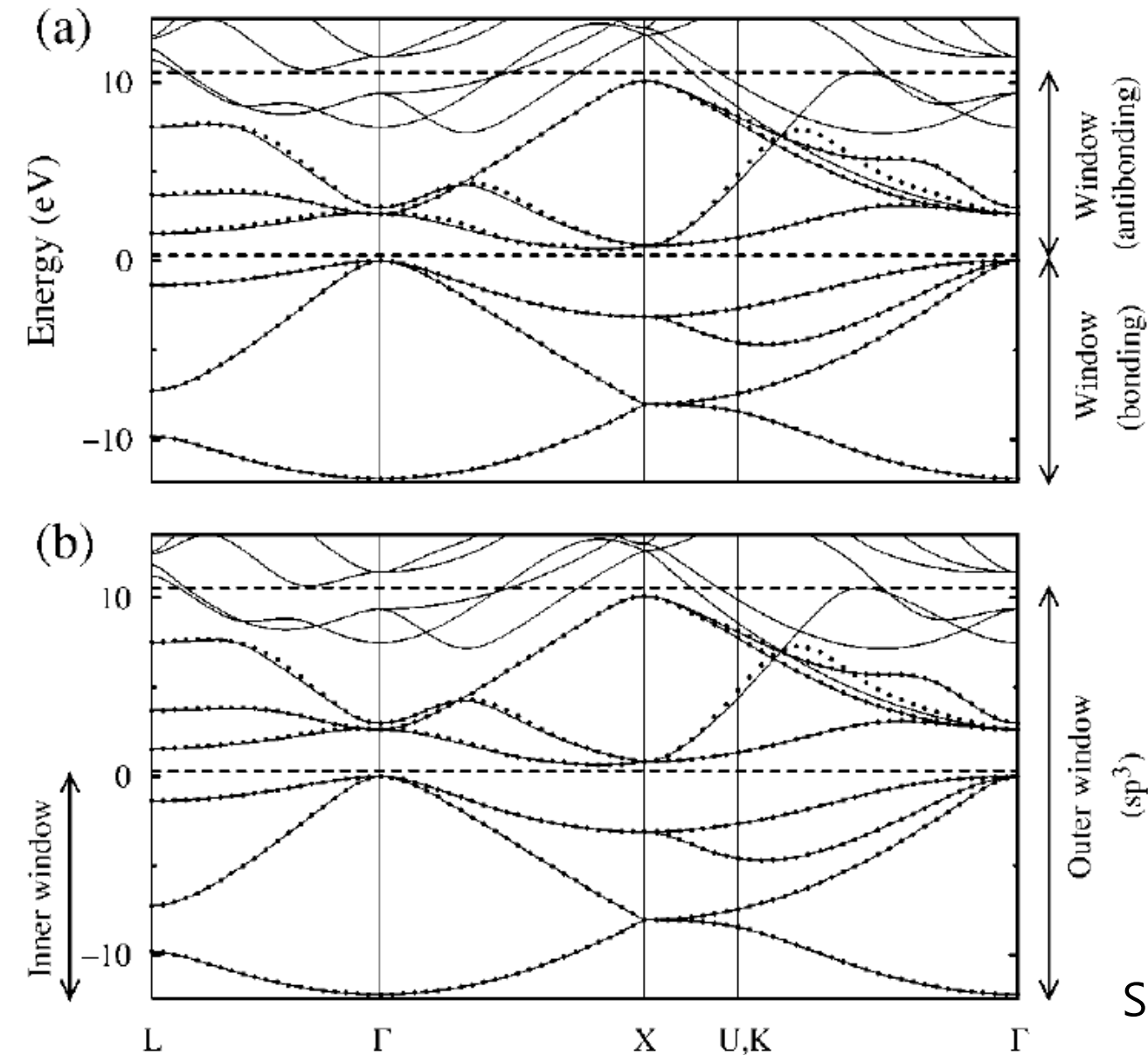
Silicon: Projection



How to select
initial guesses, and
inner/outer windows?

Souza, Marzari, Vanderbilt, PRB **65**, 035109 (2001)

Further smoothing: Maximally localized WFs in 2 steps



$$\Omega = \Omega_I + \tilde{\Omega}$$

$$\min[\Omega_I(x) + \tilde{\Omega}(x, V)] \neq \Omega_I(x_{\min}) + \min[\tilde{\Omega}(x_{\min}, V)]$$

But, in practice, the method works quite well.

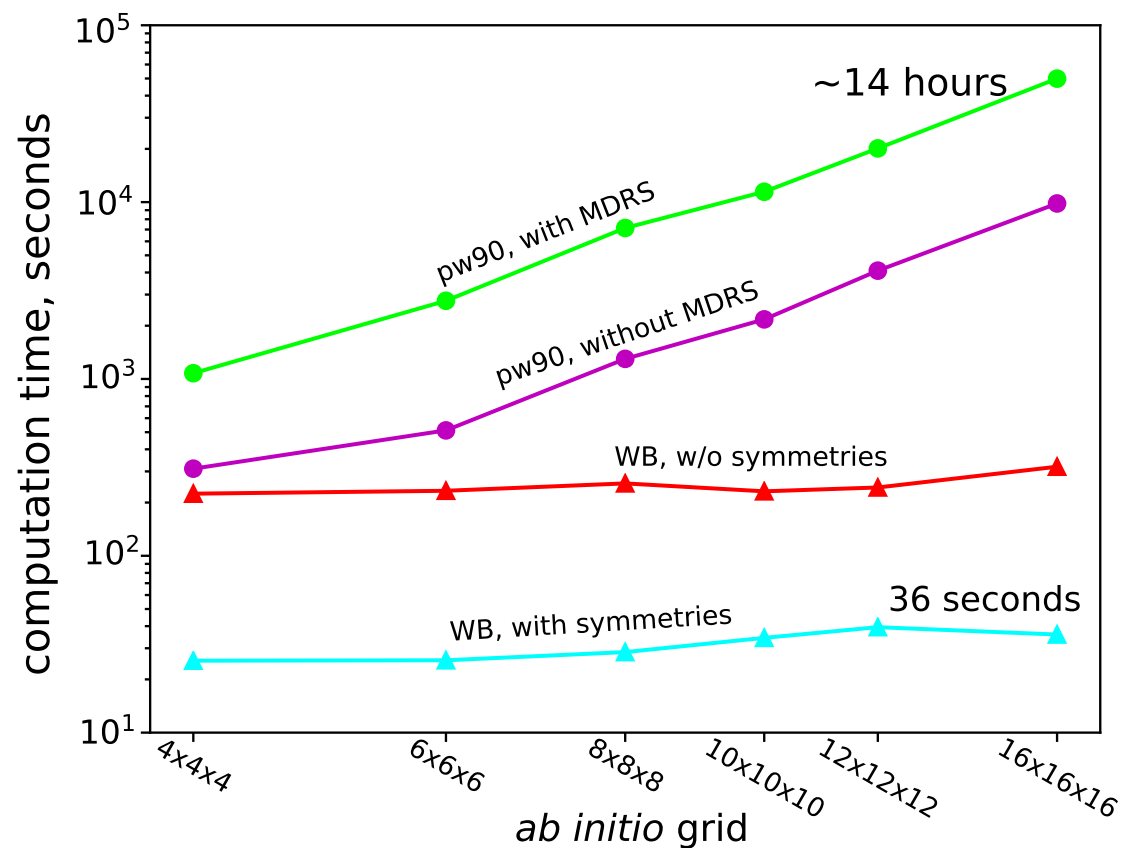
Souza, Marzari, Vanderbilt, PRB **65**, 035109 (2001)

Wannier90 & WannierBerri

https://wannier90.readthedocs.io/en/latest/user_guide/wannier90/parameters/#system-parameters

https://wannier90.readthedocs.io/en/latest/user_guide/wannier90/postproc/

<https://wannier-berri.org/index.html> Much faster, nowadays, Wannierization can also be done.



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Periodic part of Bloch sum $\langle \mathbf{r} | u_{i\mathbf{k}}^{(W)} \rangle = \frac{1}{\sqrt{N_{\mathbf{k}}}} \sum_{\mathbf{R}} e^{-i\mathbf{k} \cdot (\mathbf{r} - \mathbf{R})} w_i(\mathbf{r} - \mathbf{R}) : \quad \mathbf{r} \rightarrow \mathbf{r} + \mathbf{R}'$

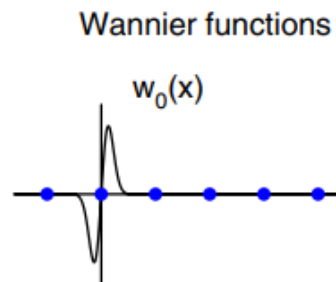
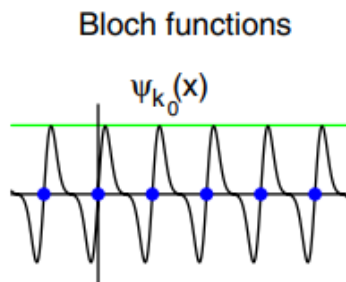
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Localized in $\mathbf{r}$

$$|w_{i\mathbf{R}}\rangle = \frac{1}{\sqrt{N_{\mathbf{k}}}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot (\hat{\mathbf{r}} - \mathbf{R})} |u_{i\mathbf{k}}^{(W)}\rangle$$

If $w_{i\mathbf{R}}$ localized at $\mathbf{r} = \mathbf{R} + \mathbf{r}_i$

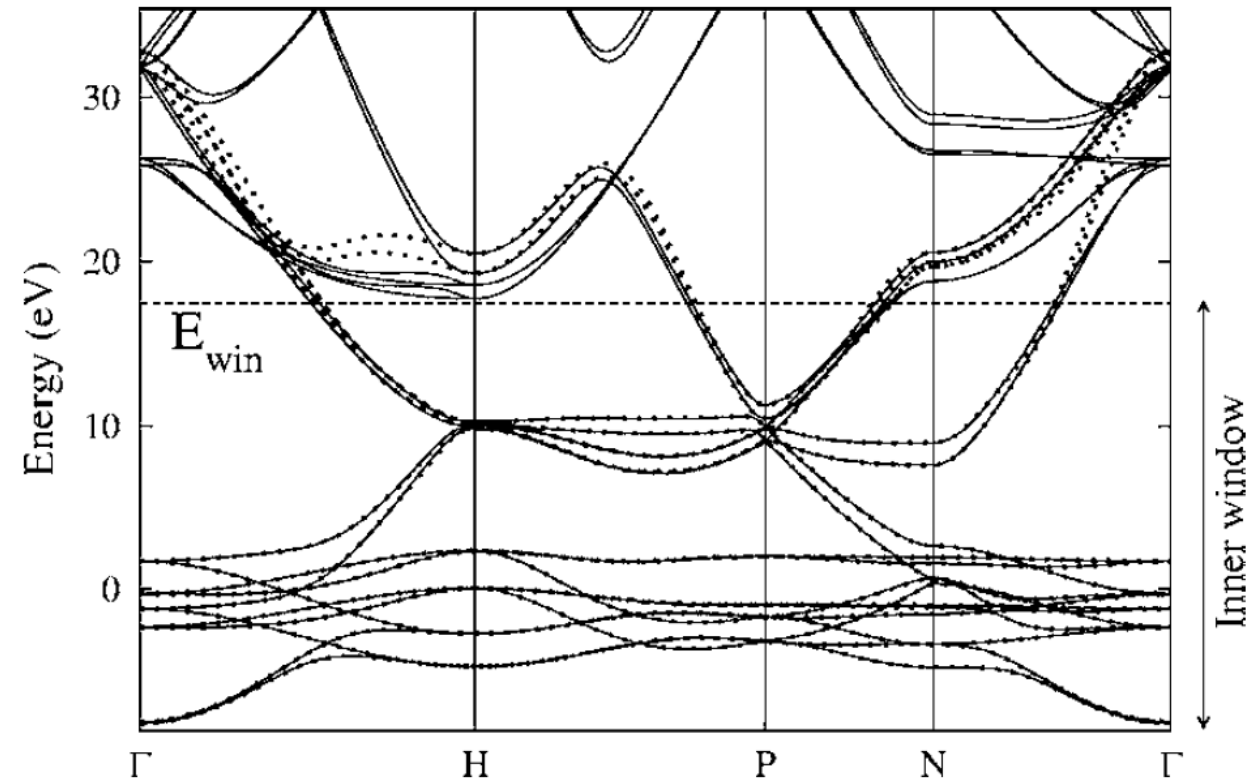


Matrix elements of a periodic operator $\hat{O}$

$$\begin{aligned}\langle \psi_{m\mathbf{k}} | \hat{O} | \psi_{n\mathbf{k}} \rangle &= \langle u_{m\mathbf{k}} | e^{-i\mathbf{k} \cdot \hat{\mathbf{r}}} \hat{O} e^{i\mathbf{k} \cdot \hat{\mathbf{r}}} | u_{n\mathbf{k}} \rangle = \langle u_{m\mathbf{k}} | \hat{O}(\mathbf{k}) | u_{n\mathbf{k}} \rangle \\ &= \sum_{ij} U_{mi\mathbf{k}} \langle u_{i\mathbf{k}}^{(W)} | \hat{O}(\mathbf{k}) | u_{j\mathbf{k}}^{(W)} \rangle U_{jn\mathbf{k}}^\dagger = \sum_{ij} U_{mi\mathbf{k}} O_{ij\mathbf{k}}^{(W)} U_{jn\mathbf{k}}^\dagger\end{aligned}$$

$$\sum_{ij} \boxed{U_{mi\mathbf{k}}} H_{ij\mathbf{k}}^{(W)} \boxed{U_{jn\mathbf{k}}^\dagger} = \delta_{mn} \boxed{\varepsilon_{m\mathbf{k}}}$$

Meaningful only if within the inner energy window



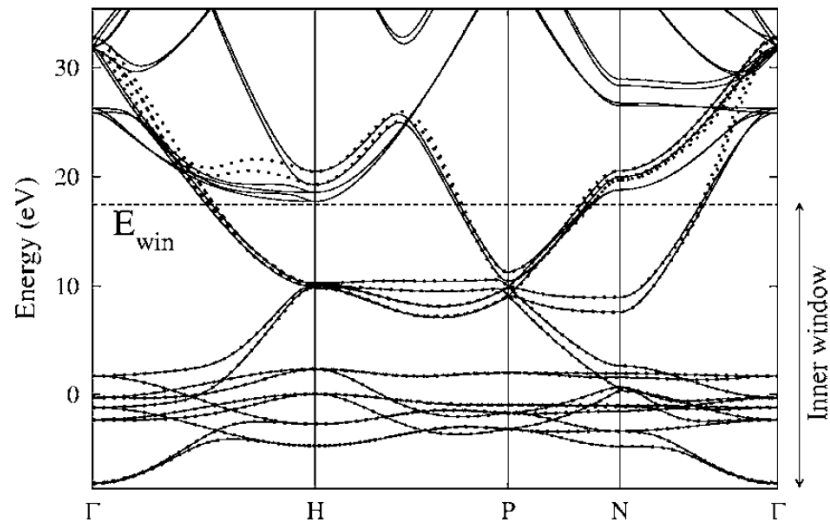
Bcc iron, Wang et al., PRB **74**, 195118 (2006)

Wannier interpolation

$$O_{ijk}^{(W)} = \frac{1}{N_{\mathbf{k}}} \sum_{\mathbf{R}', \mathbf{R}} e^{i\mathbf{k} \cdot (\mathbf{R} - \mathbf{R}')} \langle w_{i\mathbf{R}'} | \hat{O} | w_{j\mathbf{R}} \rangle = \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot \mathbf{R}} \langle w_{i\mathbf{0}} | \hat{O} | w_{j\mathbf{R}} \rangle$$

Localized in $\mathbf{R}$

Fine $\mathbf{k}$ mesh



$$\langle w_{i\mathbf{0}} | \hat{O} | w_{j\mathbf{R}} \rangle = \frac{1}{N_{\mathbf{k}}} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{R}} O_{ijk}^{(W)}$$

Coarse $\mathbf{k}$ mesh
(Smooth in $\mathbf{k}$)

$$O_{ijk}^{(W)} = \sum_{m=1}^{\tilde{N}_{\mathbf{k}}} \sum_{n=1}^{\tilde{N}_{\mathbf{k}}} V_{im\mathbf{k}}^\dagger O_{mn\mathbf{k}} V_{nj\mathbf{k}}$$

Ab initio band to smooth Wannier gauge

Ab initio matrix element

$$|u_{m\mathbf{k}}\rangle = \sum_i |u_{i\mathbf{k}}^{(W)}\rangle U_{im\mathbf{k}}^\dagger$$

Velocity operator

$$\langle \psi_{m\mathbf{k}} | \hat{\mathbf{v}} | \psi_{n\mathbf{k}} \rangle = \langle \psi_{m\mathbf{k}} | \frac{1}{i\hbar} [\hat{\mathbf{r}}, \hat{H}] | \psi_{n\mathbf{k}} \rangle \quad \text{Option 1: Do exactly the same thing}$$

$$\hat{\mathbf{v}}(\mathbf{k}) = e^{-i\mathbf{k}\cdot\hat{\mathbf{r}}} \frac{1}{i\hbar} [\hat{\mathbf{r}}, \hat{H}] e^{i\mathbf{k}\cdot\hat{\mathbf{r}}} = \frac{1}{\hbar} \nabla_{\mathbf{k}} \left(e^{-i\mathbf{k}\cdot\hat{\mathbf{r}}} \hat{H} e^{i\mathbf{k}\cdot\hat{\mathbf{r}}} \right) = \frac{1}{\hbar} \nabla_{\mathbf{k}} \hat{H}(\mathbf{k}) \quad \text{Option 2: Recycle}$$

$$\begin{aligned} \langle u_{m\mathbf{k}} | \hat{\mathbf{v}}(\mathbf{k}) | u_{n\mathbf{k}} \rangle &= \sum_{ij} U_{mi\mathbf{k}} \langle u_{i\mathbf{k}}^{(W)} | \frac{1}{\hbar} \nabla_{\mathbf{k}} \hat{H}(\mathbf{k}) | u_{j\mathbf{k}}^{(W)} \rangle U_{jn\mathbf{k}}^\dagger \\ &= \sum_{ij} U_{mi\mathbf{k}} \frac{1}{\hbar} \nabla_{\mathbf{k}} H_{ij\mathbf{k}}^{(W)} U_{jn\mathbf{k}}^\dagger + \frac{i}{\hbar} (\varepsilon_{m\mathbf{k}} - \varepsilon_{n\mathbf{k}}) \sum_{ij} U_{mi\mathbf{k}} \langle u_{i\mathbf{k}}^{(W)} | i \nabla_{\mathbf{k}} | u_{j\mathbf{k}}^{(W)} \rangle U_{jn\mathbf{k}}^\dagger \end{aligned}$$

$$\approx \mathbf{r}_i$$

$$H_{ij\mathbf{k}}^{(W)} = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} \langle w_{i0} | \hat{H} | w_{j\mathbf{R}} \rangle$$

$$|u_{i\mathbf{k}}^{(W)}\rangle \approx \frac{1}{\sqrt{N_{\mathbf{k}}}} e^{-i\mathbf{k}\cdot\mathbf{r}_i} \sum_{\mathbf{R}} |w_{i\mathbf{R}}\rangle$$

If $\omega_{i\mathbf{R}}$ localized at $\mathbf{r}=\mathbf{R}+\mathbf{r}_i$

$$\nabla_{\mathbf{k}} H_{ij\mathbf{k}}^{(W)} = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} (i\mathbf{R}) \langle w_{i0} | \hat{H} | w_{j\mathbf{R}} \rangle$$

Also localized in $\mathbf{R}$
Recycle

Position operator

If $w_{i\mathbf{R}}$ is strongly localized at $\mathbf{r}=\mathbf{R}+\mathbf{r}_i$

$$\approx \mathbf{r}_i$$

$$|u_{i\mathbf{k}}^{(W)}\rangle = \frac{1}{\sqrt{N_{\mathbf{k}}}} \sum_{\mathbf{R}'} e^{-i\mathbf{k}\cdot(\hat{\mathbf{r}}-\mathbf{R}')} |w_{i\mathbf{R}'}\rangle$$

$$i\nabla_{\mathbf{k}} |u_{j\mathbf{k}}^{(W)}\rangle = \frac{1}{\sqrt{N_{\mathbf{k}}}} \sum_{\mathbf{R}} e^{-i\mathbf{k}\cdot(\hat{\mathbf{r}}-\mathbf{R})} (\hat{\mathbf{r}} - \mathbf{R}) |w_{j\mathbf{R}}\rangle$$

$$\langle u_{i\mathbf{k}}^{(W)} | i\nabla_{\mathbf{k}} | u_{j\mathbf{k}}^{(W)} \rangle = \frac{1}{N_{\mathbf{k}}} \sum_{\mathbf{R}', \mathbf{R}} e^{i\mathbf{k}\cdot(\mathbf{R}-\mathbf{R}')} \langle w_{i\mathbf{R}'} | (\hat{\mathbf{r}} - \mathbf{R}) | w_{j\mathbf{R}} \rangle$$

Fine $\mathbf{k}$ mesh

$$= \frac{1}{N_{\mathbf{k}}} \sum_{\mathbf{R}', \mathbf{R}} e^{i\mathbf{k}\cdot(\mathbf{R}-\mathbf{R}')} \langle w_{i\mathbf{R}'-\mathbf{R}'} | [\hat{\mathbf{r}} - (\mathbf{R} - \mathbf{R}')] | w_{j\mathbf{R}-\mathbf{R}'} \rangle$$

$$\langle w_{i0} | \mathbf{R} | w_{j\mathbf{R}} \rangle = \mathbf{R} \delta_{ij} \delta_{\mathbf{R}0} = \mathbf{0}$$

$$= \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} \langle w_{i0} | (\hat{\mathbf{r}} - \mathbf{R}) | w_{j\mathbf{R}} \rangle = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} \langle w_{i0} | \hat{\mathbf{r}} | w_{j\mathbf{R}} \rangle \quad \text{Recycle}$$

$$\mathbf{r}_{ij\mathbf{R}} = \langle w_{i0} | \hat{\mathbf{r}} | w_{j\mathbf{R}} \rangle = \frac{1}{N_{\mathbf{k}}} \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{R}} \langle u_{i\mathbf{k}}^{(W)} | i\nabla_{\mathbf{k}} | u_{j\mathbf{k}}^{(W)} \rangle$$

Coarse $\mathbf{k}$ mesh

Position matrix elements

$$\mathbf{r}_{ij\mathbf{R}} = \langle w_{i\mathbf{0}} | \hat{\mathbf{r}} | w_{j\mathbf{R}} \rangle$$

Diagonal part: Wannier center $\mathbf{r}_i = \langle w_{i\mathbf{0}} | \hat{\mathbf{r}} | w_{i\mathbf{0}} \rangle$

Usage

- Transport
- Berry-related quantities
- Orbital magnetization
- Optical responses

Finite difference method

$$\nabla_{\mathbf{k}} f(\mathbf{k}) \approx \sum_{\mathbf{b}} c_{|\mathbf{b}|} \mathbf{b} f(\mathbf{k} + \mathbf{b}) \quad \sum_{\mathbf{b}} c_{|\mathbf{b}|} b^{\alpha} b^{\beta} = \delta_{\alpha\beta}$$

$\Delta \mathbf{k}$ for coarse $\mathbf{k}$ mesh $\sim \frac{1}{N_k}$

e.g. 1D, 1st-order case

$$\frac{df}{dx} \approx \frac{f(x+h) - f(x-h)}{2h} = \sum_{b \in \{h, -h\}} \left[\frac{1}{2b^2} \right] \cdot b \cdot f(x+b)$$

$\sum_{\mathbf{b}} c_{|\mathbf{b}|} \mathbf{b} f(\mathbf{k} + \mathbf{b})$

(Arrows point from the boxed term $\frac{1}{2b^2}$ to the coefficient $c_{|\mathbf{b}|}$ and from the b term to the vector $\mathbf{b}$ in the general formula above.)

Position operator: Finite difference approximation

$$\mathbf{r}_{ij\mathbf{R}} = \langle w_{i0} | \hat{\mathbf{r}} | w_{j\mathbf{R}} \rangle$$

$$= \frac{1}{N_{\mathbf{k}}} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{R}} \langle u_{i\mathbf{k}}^{(W)} | i \nabla_{\mathbf{k}} | u_{j\mathbf{k}}^{(W)} \rangle$$

$$\approx \frac{i}{N_{\mathbf{k}}} \sum_{\mathbf{k}\mathbf{b}} c_{|\mathbf{b}|} \mathbf{b} e^{-i\mathbf{k} \cdot \mathbf{R}} \langle u_{i\mathbf{k}}^{(W)} | u_{j\mathbf{k}+\mathbf{b}}^{(W)} \rangle$$

$$= \frac{i}{N_{\mathbf{k}}^2} \sum_{\mathbf{k}\mathbf{b}\mathbf{R}''\mathbf{R}'} c_{|\mathbf{b}|} \mathbf{b} e^{-i\mathbf{k} \cdot [\mathbf{R} - (\mathbf{R}' - \mathbf{R}'')]} \langle w_{i\mathbf{R}''} | e^{-i\mathbf{b} \cdot (\hat{\mathbf{r}} - \mathbf{R}')} | w_{i\mathbf{R}'} \rangle$$

$$= \frac{i}{N_{\mathbf{k}}} \sum_{\mathbf{k}\mathbf{b}\mathbf{R}'} c_{|\mathbf{b}|} \mathbf{b} e^{-i\mathbf{k} \cdot (\mathbf{R} - \mathbf{R}')} \langle w_{i0} | e^{-i\mathbf{b} \cdot (\hat{\mathbf{r}} - \mathbf{R}')} | w_{i\mathbf{R}'} \rangle$$

$$= i \sum_{\mathbf{b}} c_{|\mathbf{b}|} \mathbf{b} \langle w_{i0} | e^{-i\mathbf{b} \cdot (\hat{\mathbf{r}} - \mathbf{R})} | w_{i\mathbf{R}} \rangle$$

$$|u_{i\mathbf{k}}^{(W)}\rangle = \frac{1}{\sqrt{N_{\mathbf{k}}}} \sum_{\mathbf{R}''} e^{-i\mathbf{k} \cdot (\hat{\mathbf{r}} - \mathbf{R}'')} |w_{i\mathbf{R}''}\rangle$$

$$|u_{j\mathbf{k}+\mathbf{b}}^{(W)}\rangle = \frac{1}{\sqrt{N_{\mathbf{k}}}} \sum_{\mathbf{R}'} e^{-i(\mathbf{k}+\mathbf{b}) \cdot (\hat{\mathbf{r}} - \mathbf{R}')} |w_{i\mathbf{R}'}\rangle$$

What does this amount to?

Position operator: Finite difference approximation

$$\begin{aligned}
 \langle w_{i0} | \hat{\mathbf{r}} | w_{j\mathbf{R}} \rangle &\approx \frac{i}{N_{\mathbf{k}}} \sum_{\mathbf{k}\mathbf{b}} c_{|\mathbf{b}|} \mathbf{b} e^{-i\mathbf{k}\cdot\mathbf{R}} \langle u_{i\mathbf{k}}^{(W)} | u_{j\mathbf{k}+\mathbf{b}}^{(W)} \rangle \\
 &= i \sum_{\mathbf{b}} c_{|\mathbf{b}|} \mathbf{b} \langle w_{i0} | e^{-i\mathbf{b}\cdot(\hat{\mathbf{r}}-\mathbf{R})} | w_{j\mathbf{R}} \rangle \\
 &\approx i \sum_{\mathbf{b}} c_{|\mathbf{b}|} \mathbf{b} \langle w_{i0} | [1 - i\mathbf{b} \cdot (\hat{\mathbf{r}} - \mathbf{R})] | w_{j\mathbf{R}} \rangle = \langle w_{i0} | \hat{\mathbf{r}} | w_{j\mathbf{R}} \rangle
 \end{aligned}$$

$$c_{|\mathbf{b}|} \mathbf{b} + c_{|-\mathbf{b}|} (-\mathbf{b}) = 0$$

$$\sum_{\mathbf{b}} c_{|\mathbf{b}|} b^{\alpha} b^{\beta} = \delta_{\alpha\beta}$$

$$\begin{aligned} \langle w_{i0} | \mathbf{R} | w_{j\mathbf{R}} \rangle \\ = \mathbf{R} \delta_{ij} \delta_{\mathbf{R}0} = 0 \end{aligned}$$

Good approximation if $\mathbf{b} \rightarrow 0$, i.e., the coarse $\mathbf{k}$ mesh is very dense, or, Wannier functions are very localized **and** Wannier centers $\mathbf{r}_i$ and $\mathbf{r}_j + \mathbf{R}$ are both very close to $\mathbf{R}$ **(But, the latter condition also depends on the origin.)**

Translationally equivariant finite-difference formalism

$$\mathbf{r}_{ij\mathbf{R}} = \langle w_{i0} | \hat{\mathbf{r}} | w_{j\mathbf{R}} \rangle \quad \mathbf{r}'_i = \mathbf{r}_i + \mathbf{d}, \quad \mathbf{r}'_j = \mathbf{r}_j + \mathbf{d}$$

$$\mathbf{r}'_{ij\mathbf{R}} = \mathbf{r}_{ij\mathbf{R}} + \mathbf{d} \delta_{ij} \delta_{\mathbf{R}0}$$

Translational equivariance
(due to change in the origin)
Not satisfied in simple finite difference (S-FD)

$$\bar{\mathbf{r}}_{ij\mathbf{R}} = \frac{\mathbf{r}_i + \mathbf{r}_j + \mathbf{R}}{2}$$

$$\bar{\mathbf{r}}'_{ij\mathbf{R}} = \bar{\mathbf{r}}_{ij\mathbf{R}} + \mathbf{d}$$

$$\hat{\mathbf{r}} \approx \hat{\mathbf{r}}_{ij\mathbf{R}}^{\text{TEFD}} = \bar{\mathbf{r}}_{ij\mathbf{R}} + i \sum_{\mathbf{b}} c_{|\mathbf{b}|} \mathbf{b} e^{-i\mathbf{b} \cdot (\hat{\mathbf{r}} - \bar{\mathbf{r}}_{ij\mathbf{R}})}$$

Good approx. near the average of
the two Wannier centers

$$\mathbf{r}_{ij\mathbf{R}}^{\text{TEFD}} = \langle w_{i0} | \hat{\mathbf{r}}_{ij\mathbf{R}}^{\text{TEFD}} | w_{j\mathbf{R}} \rangle$$

independent of the origin
(translational equivariance)

$$= \bar{\mathbf{r}}_{ij\mathbf{R}} \delta_{ij} \delta_{\mathbf{R}0} + i \sum_{\mathbf{b}} c_{|\mathbf{b}|} \mathbf{b} \langle w_{i0} | e^{-i\mathbf{b} \cdot (\hat{\mathbf{r}} - \bar{\mathbf{r}}_{ij\mathbf{R}})} | w_{j\mathbf{R}} \rangle$$

Translationally equivariant finite-difference formalism

$$|w_{i0}\rangle = \frac{1}{\sqrt{N_{\mathbf{k}}}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\hat{\mathbf{r}}} |u_{i\mathbf{k}}^{(W)}\rangle$$

$$|w_{j\mathbf{R}}\rangle = \frac{1}{\sqrt{N_{\mathbf{k}}}} \sum_{\mathbf{k}'} e^{i\mathbf{k}'\cdot(\hat{\mathbf{r}}-\mathbf{R})} |u_{j\mathbf{k}'}^{(W)}\rangle$$

$$\begin{aligned} \mathbf{r}_{ij\mathbf{R}}^{\text{TEFD}} &= \bar{\mathbf{r}}_{ij\mathbf{R}} \delta_{ij} \delta_{\mathbf{R}0} + i \sum_{\mathbf{b}} c_{|\mathbf{b}|} \mathbf{b} \langle w_{i0} | e^{-i\mathbf{b}\cdot(\hat{\mathbf{r}}-\bar{\mathbf{r}}_{ij\mathbf{R}})} | w_{j\mathbf{R}} \rangle \\ &= \mathbf{r}_i \delta_{ij} \delta_{\mathbf{R}0} + \frac{i}{N_{\mathbf{k}}} \sum_{\mathbf{k}\mathbf{k}'\mathbf{b}} c_{|\mathbf{b}|} \mathbf{b} e^{i\mathbf{b}\cdot\bar{\mathbf{r}}_{ij\mathbf{R}}} \langle u_{i\mathbf{k}}^{(W)} | e^{-i(\mathbf{k}+\mathbf{b}-\mathbf{k}')\cdot\hat{\mathbf{r}}} | u_{j\mathbf{k}'}^{(W)} \rangle e^{-i\mathbf{k}'\cdot\mathbf{R}} \\ &= \frac{i}{N_{\mathbf{k}}} \sum_{\mathbf{k}\mathbf{b}} c_{|\mathbf{b}|} \mathbf{b} e^{-i\mathbf{k}\cdot\mathbf{R}} \langle u_{i\mathbf{k}}^{(W)} | u_{j\mathbf{k}+\mathbf{b}}^{(W)} \rangle \boxed{e^{i\mathbf{b}\cdot(\bar{\mathbf{r}}_{ij\mathbf{R}}-\mathbf{R})}} + \boxed{\mathbf{r}_i \delta_{ij} \delta_{\mathbf{R}0}} \end{aligned}$$

Extra phase
Extra term

Composite operators

Orbital magnetization

$$i\langle u_{i\mathbf{k}}^{(W)} | \hat{H}_{\mathbf{k}} | \partial^\alpha u_{j\mathbf{k}}^{(W)} \rangle \text{ and } \langle \partial^\alpha u_{i\mathbf{k}}^{(W)} | \hat{H}_{\mathbf{k}} | \partial^\beta u_{j\mathbf{k}}^{(W)} \rangle$$

Lopez, Vanderbilt, Thonhauser, and Souza, PRB **74**, 195118 (2006)

Spin-Hall conductivity

$$i\langle u_{i\mathbf{k}}^{(W)} | \hat{s}^\gamma | \partial^\alpha u_{j\mathbf{k}}^{(W)} \rangle \text{ and } i\langle u_{i\mathbf{k}}^{(W)} | \hat{s}^\gamma \hat{H}_{\mathbf{k}} | \partial^\alpha u_{j\mathbf{k}}^{(W)} \rangle$$

Ryoo, Park, and Souza, PRB **99**, 235113 (2019)

Lihm, Ghim, Hong, and Park, arXiv:2604.22614 (2026)

General case

$$B_{ij\mathbf{R}}^\alpha[\hat{O}] = \langle w_{i\mathbf{0}} | \hat{O}(\hat{r} - R)^\alpha | w_{j\mathbf{R}} \rangle$$

$$C_{ij\mathbf{R}}^{\alpha\beta}[\hat{O}] = \langle w_{i\mathbf{0}} | \hat{r}^\alpha \hat{O}(\hat{r} - R)^\beta | w_{j\mathbf{R}} \rangle$$

Finite-difference formalism

$$B_{ij}^{(\mathbf{k}, \mathbf{b})}[\hat{O}] = \langle u_{i\mathbf{k}}^{(W)} | \hat{O}_{\mathbf{k}} | u_{j\mathbf{k}+\mathbf{b}}^{(W)} \rangle$$

$$C_{ij}^{(\mathbf{k}, \mathbf{b}_1, \mathbf{b}_2)}[\hat{O}] = \langle u_{i\mathbf{k}+\mathbf{b}_1}^{(W)} | \hat{O}_{\mathbf{k}} | u_{j\mathbf{k}+\mathbf{b}_2}^{(W)} \rangle$$

$$B_{ij\mathbf{R}}^{\alpha(\text{S-FD})}[\hat{O}] = i \sum_{\mathbf{b}} c_{|\mathbf{b}|} b^{\alpha} \langle w_{i0} | \hat{O} e^{-i\mathbf{b} \cdot (\hat{\mathbf{r}} - \mathbf{R})} | w_{j\mathbf{R}} \rangle$$

$$= \frac{i}{N_{\mathbf{k}}} \sum_{\mathbf{bk}} c_{|\mathbf{b}|} b^{\alpha} e^{-i\mathbf{k} \cdot \mathbf{R}} B_{ij}^{(\mathbf{k}, \mathbf{b})}[\hat{O}]$$

$$C_{ij\mathbf{R}}^{\alpha\beta(\text{S-FD})}[\hat{O}] = \sum_{\mathbf{b}_1 \mathbf{b}_2} c_{|\mathbf{b}_1|} c_{|\mathbf{b}_2|} b_1^{\alpha} b_2^{\beta} \langle w_{i0} | e^{i\mathbf{b}_1 \cdot \hat{\mathbf{r}}} \hat{O} e^{-i\mathbf{b}_2 \cdot (\hat{\mathbf{r}} - \mathbf{R})} | w_{j\mathbf{R}} \rangle$$

$$= \frac{1}{N_{\mathbf{k}}} \sum_{\mathbf{b}_1 \mathbf{b}_2 \mathbf{k}} c_{|\mathbf{b}_1|} c_{|\mathbf{b}_2|} b_1^{\alpha} b_2^{\beta} e^{-i\mathbf{k} \cdot \mathbf{R}} C_{ij}^{(\mathbf{k}, \mathbf{b}_1, \mathbf{b}_2)}[\hat{O}]$$

Translationally equivariant finite-difference formalism

$$B_{ij}^{(\mathbf{k}, \mathbf{b})}[\hat{O}] = \langle u_{i\mathbf{k}}^{(W)} | \hat{O}_{\mathbf{k}} | u_{j\mathbf{k}+\mathbf{b}}^{(W)} \rangle$$

$$C_{ij}^{(\mathbf{k}, \mathbf{b}_1, \mathbf{b}_2)}[\hat{O}] = \langle u_{i\mathbf{k}+\mathbf{b}_1}^{(W)} | \hat{O}_{\mathbf{k}} | u_{j\mathbf{k}+\mathbf{b}_2}^{(W)} \rangle$$

$$B_{ij\mathbf{R}}^{\alpha(\text{TEFD})}[\hat{O}] = \frac{i}{N_{\mathbf{k}}} \sum_{\mathbf{b}\mathbf{k}} c_{|\mathbf{b}|} b^{\alpha} e^{-i\mathbf{k}\cdot\mathbf{R}} B_{ij}^{(\mathbf{k}, \mathbf{b})}[\hat{O}] \boxed{e^{i\mathbf{b}\cdot(\bar{\mathbf{r}}_{ij\mathbf{R}} - \mathbf{R})}} + \boxed{(\bar{r}_{ij\mathbf{R}} - R)^{\alpha} O_{ij\mathbf{R}}}$$

Extra phase

Extra terms

$$C_{ij\mathbf{R}}^{\alpha\beta(\text{TEFD})}[\hat{O}] = \frac{1}{N_{\mathbf{k}}} \sum_{\mathbf{b}_1 \mathbf{b}_2 \mathbf{k}} c_{|\mathbf{b}_1|} c_{|\mathbf{b}_2|} b_1^{\alpha} b_2^{\beta} e^{-i\mathbf{k}\cdot\mathbf{R}} C_{ij}^{(\mathbf{k}, \mathbf{b}_1, \mathbf{b}_2)}[\hat{O}] \boxed{e^{-i\mathbf{b}_1\cdot\bar{\mathbf{r}}_{ij\mathbf{R}} + i\mathbf{b}_2\cdot(\bar{\mathbf{r}}_{ij\mathbf{R}} - \mathbf{R})}}$$

$$+ \boxed{\bar{r}_{ij\mathbf{R}}^{\alpha} B_{ij\mathbf{R}}^{\beta(\text{TEFD})}[\hat{O}] + B_{ji-\mathbf{R}}^{\alpha*(\text{TEFD})}[\hat{O}^{\dagger}] (\bar{r}_{ij\mathbf{R}} - R)^{\beta} - \bar{r}_{ij\mathbf{R}}^{\alpha} (\bar{r}_{ij\mathbf{R}} - R)^{\beta} O_{ij\mathbf{R}}}$$

Finite difference method

$$\nabla_{\mathbf{k}} f(\mathbf{k}) \approx \sum_{\mathbf{b}} c_{|\mathbf{b}|} \mathbf{b} f(\mathbf{k} + \mathbf{b}) \quad \sum_{\mathbf{b}} c_{|\mathbf{b}|} b^{\alpha} b^{\beta} = \delta_{\alpha\beta}$$

$\Delta \mathbf{k}$ for coarse $\mathbf{k}$ mesh $\sim \frac{1}{N_k}$

e.g. 1D, 1st-order case

$$\frac{df}{dx} \approx \frac{f(x+h) - f(x-h)}{2h} = \sum_{b \in \{h, -h\}} \left[\frac{1}{2b^2} \right] \cdot b \cdot f(x+b)$$

$\sum_{\mathbf{b}} c_{|\mathbf{b}|} \mathbf{b} f(\mathbf{k} + \mathbf{b})$

Higher-order finite difference method

One-dimensional case

1st-order

$$\frac{\partial f}{\partial x} = \frac{f(x+h) - f(x-h)}{2h} + \boxed{O(h^2)}$$


2nd-order

$$\frac{\partial f}{\partial x} = \frac{2f(x+h) - 2f(x-h)}{3h} + \frac{-f(x+2h) + f(x-2h)}{12h} + \boxed{O(h^4)}$$


3rd-order

$$\frac{\partial f}{\partial x} = \frac{3f(x+h) - 3f(x-h)}{4h} + \frac{-f(x+2h) + f(x-2h)}{15h} + \frac{f(x+3h) - f(x-3h)}{60h} + \boxed{O(h^6)}$$


1st neighbor

2nd neighbor

3rd neighbor

Higher-order finite difference method

Three-dimensional case

1st-order

$$\nabla f = \sum_{\mathbf{b}} \underline{c_{|\mathbf{b}|}^{(1)} \mathbf{b} f(\mathbf{k} + \mathbf{b})} + O(b^2)$$

2nd-order

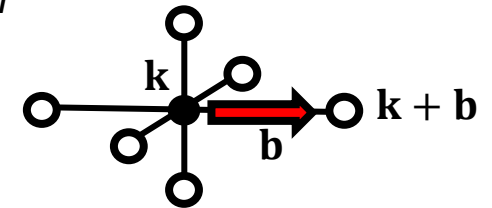
$$\nabla f = \sum_{\mathbf{b}} \underline{c_{|\mathbf{b}|}^{(2)} \mathbf{b} f(\mathbf{k} + \mathbf{b})} + \sum_{\mathbf{b}} \underline{c_{|2\mathbf{b}|}^{(2)} 2\mathbf{b} f(\mathbf{k} + 2\mathbf{b})} + O(b^4)$$

3rd-order

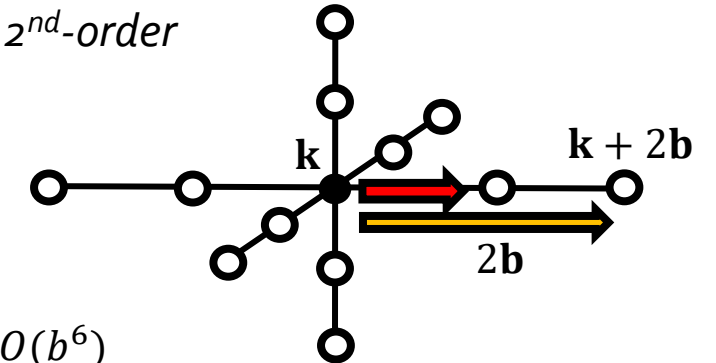
$$\nabla f = \sum_{\mathbf{b}} \underline{c_{|\mathbf{b}|}^{(3)} \mathbf{b} f(\mathbf{k} + \mathbf{b})} + \sum_{\mathbf{b}} \underline{c_{|2\mathbf{b}|}^{(3)} 2\mathbf{b} f(\mathbf{k} + 2\mathbf{b})} + \sum_{\mathbf{b}} \underline{c_{|3\mathbf{b}|}^{(3)} 3\mathbf{b} f(\mathbf{k} + 3\mathbf{b})} + O(b^6)$$

1st neighbor 2nd neighbor 3rd neighbor

1st-order



2nd-order



⋮

Higher-order finite difference method

- N^{th} -order, we found:

$$\nabla_{\alpha} f(\mathbf{k}) \approx \nabla_{\alpha}^{\text{Nth-FD}} f(\mathbf{k}) = \sum_{m=1}^N \sum_{\mathbf{b}} c_{|m\mathbf{b}|}^{(N)} m b^{\alpha} f(\mathbf{k} + m\mathbf{b})$$

using “multiples” of $\mathbf{b}$ vectors, $m\mathbf{b}$.

$$c_{|m\mathbf{b}|}^{(N)} = c_{|\mathbf{b}|}^{(1)} \frac{1}{m^2} \prod_{n \neq m}^N \frac{n^2}{n^2 - m^2}$$

- Error $\sim O(b^{2N}) = O(N_k^{-2N})$

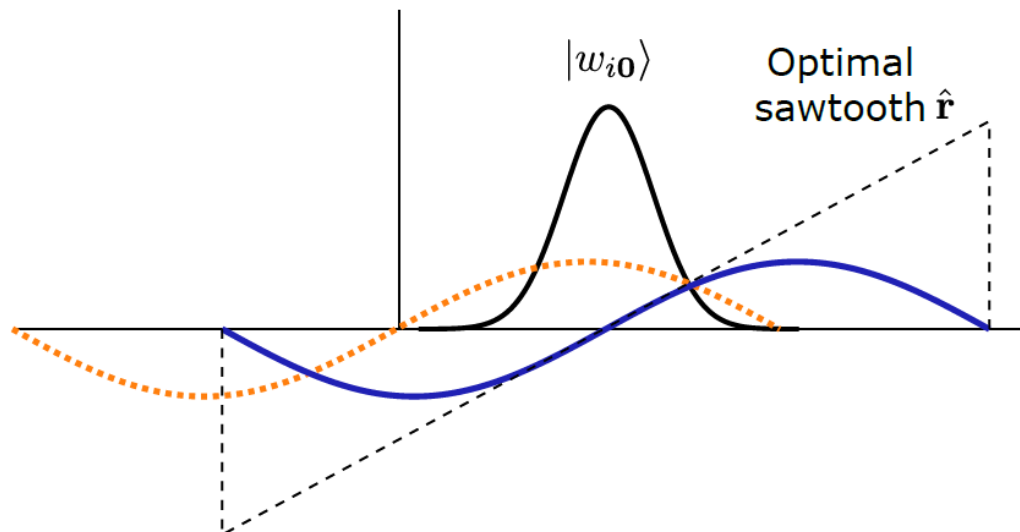
Higher-order finite difference method (S-FD for simplicity)

$$\begin{aligned} & i \sum_{\mathbf{b}} c_{|\mathbf{b}|} \mathbf{b} \langle w_{i\mathbf{0}} | e^{-i\mathbf{b} \cdot (\hat{\mathbf{r}} - \mathbf{R})} | w_{j\mathbf{R}} \rangle \\ & \approx i \sum_{\mathbf{b}} c_{|\mathbf{b}|} \mathbf{b} \langle w_{i\mathbf{0}} | [1 - i\mathbf{b} \cdot (\hat{\mathbf{r}} - \mathbf{R})] | w_{j\mathbf{R}} \rangle = \langle w_{i\mathbf{0}} | \hat{\mathbf{r}} | w_{j\mathbf{R}} \rangle \\ & \sum_{\mathbf{b}} c_{|\mathbf{b}|} b^\alpha b^\beta = \delta_{\alpha\beta} \end{aligned}$$

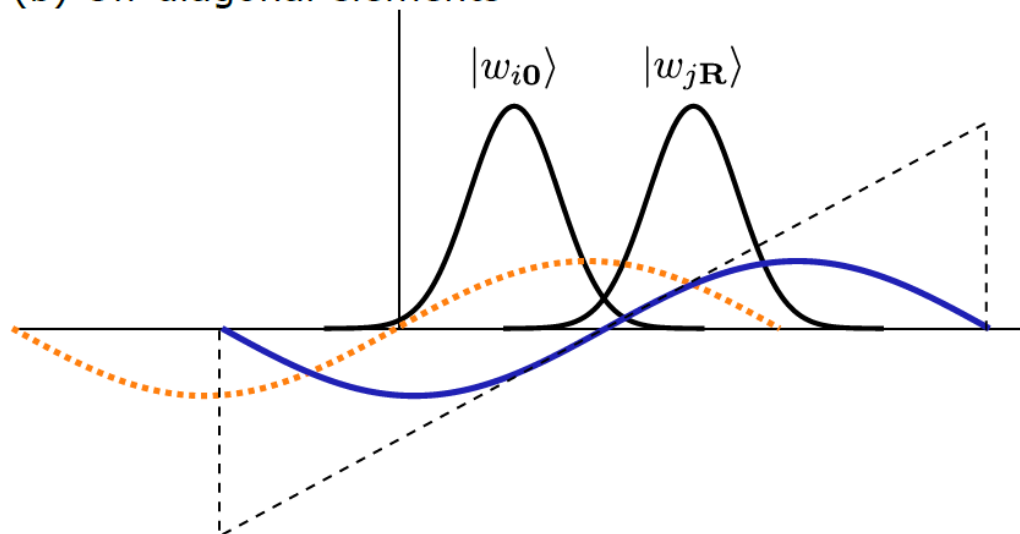
Add more sinusoidal waves ($\mathbf{b}$, $2\mathbf{b}$, ..., $N\mathbf{b}$) to mimic the linear operator by removing higher-order errors

Translational equivariance

(a) Diagonal elements (Wannier centers)



(b) Off-diagonal elements

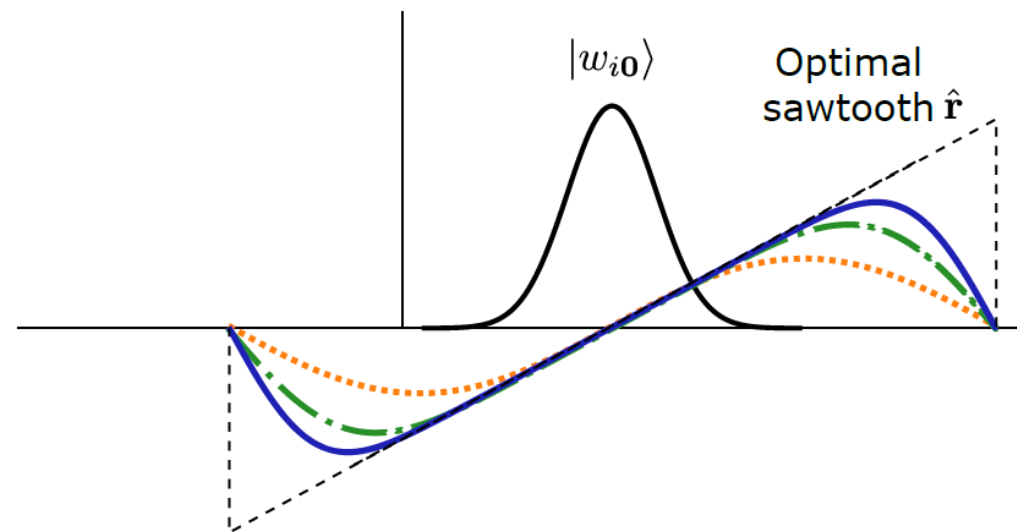


..... Simple finite difference

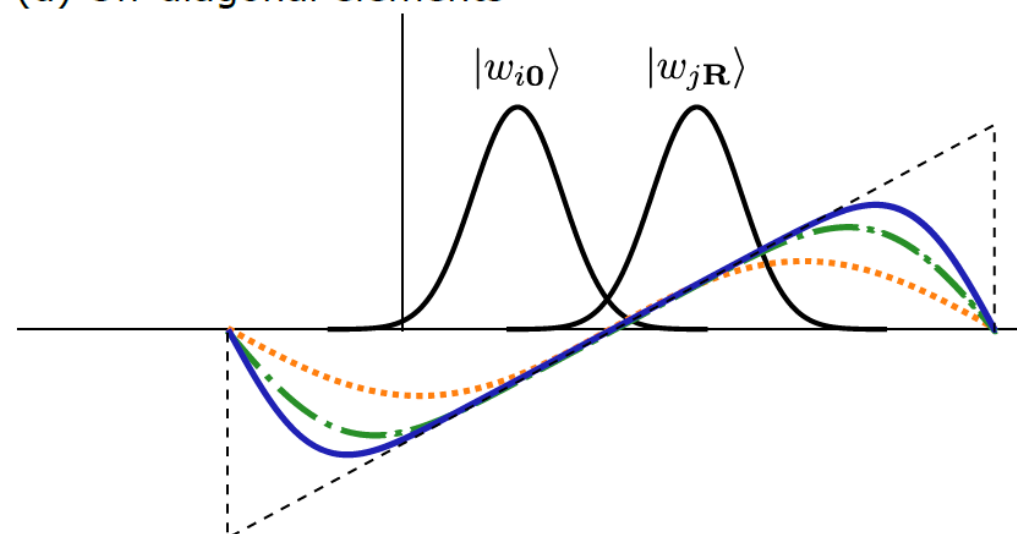
— Translationally-equivariant finite difference

Higher-order finite difference

(c) Diagonal elements (Wannier centers)



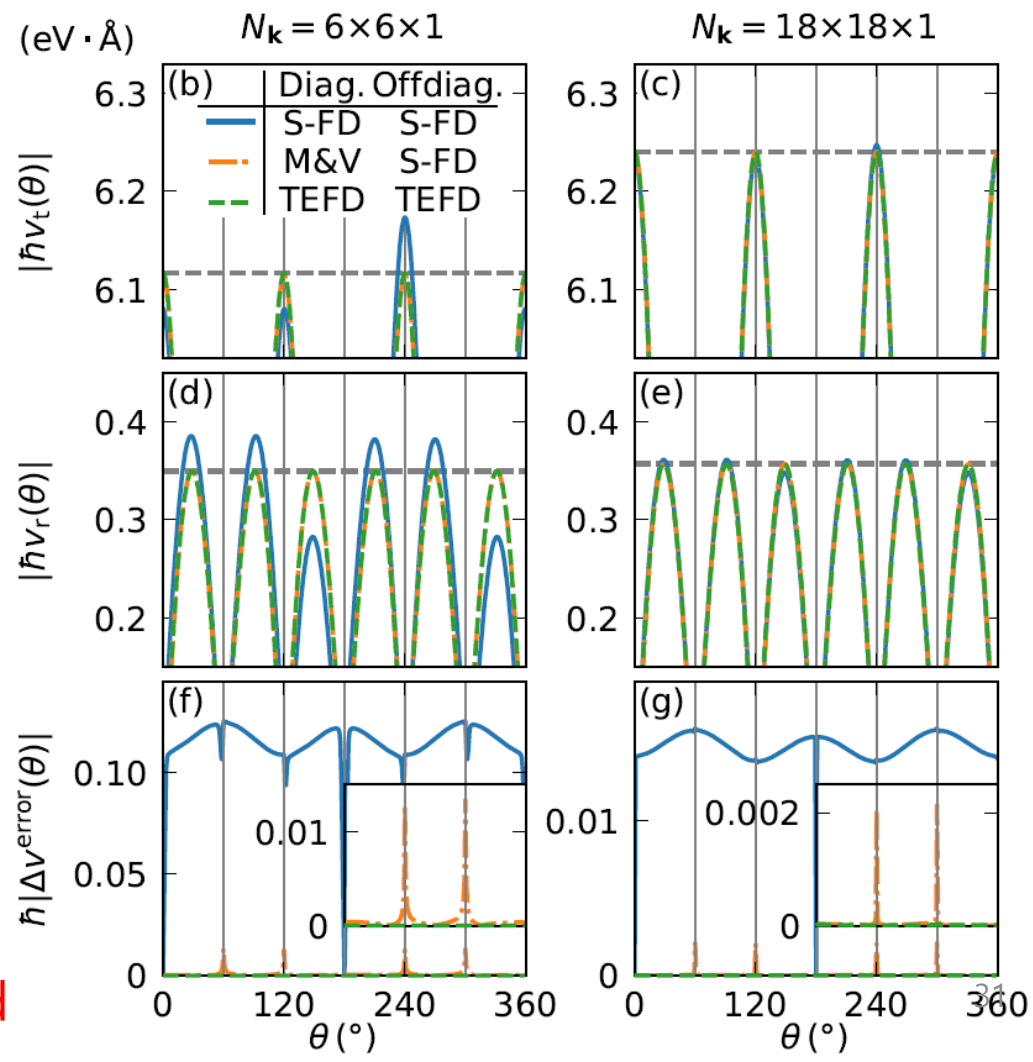
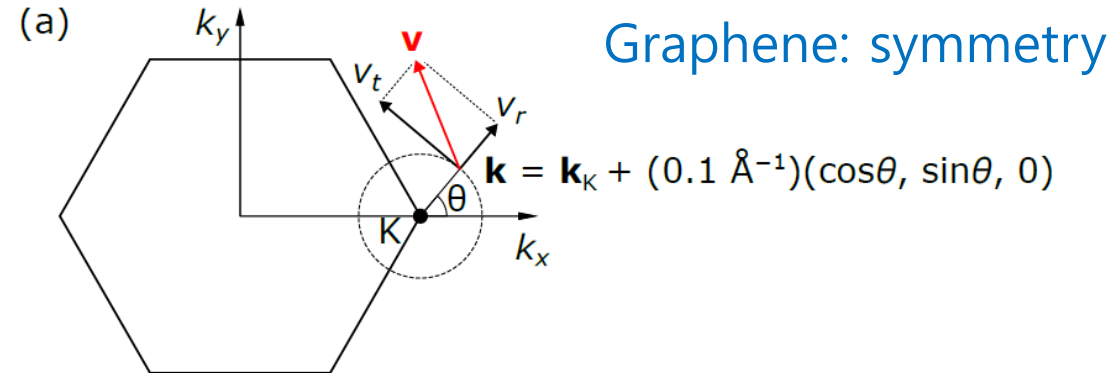
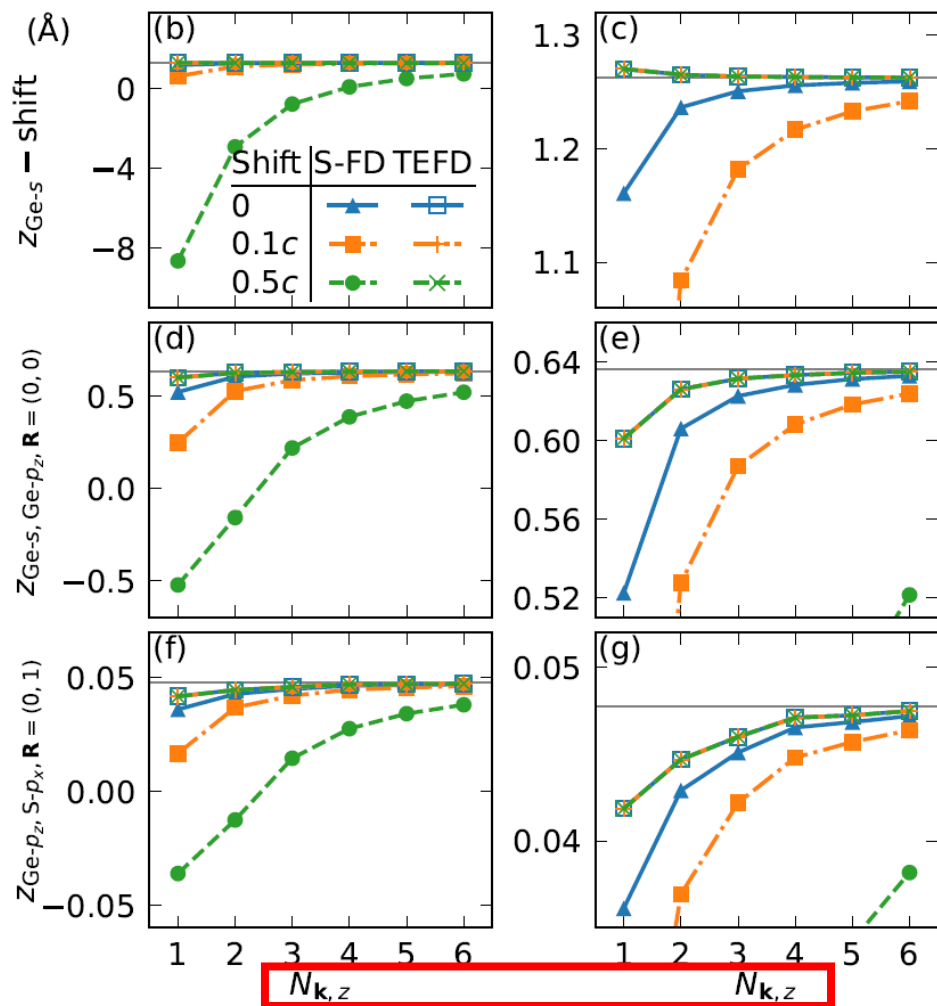
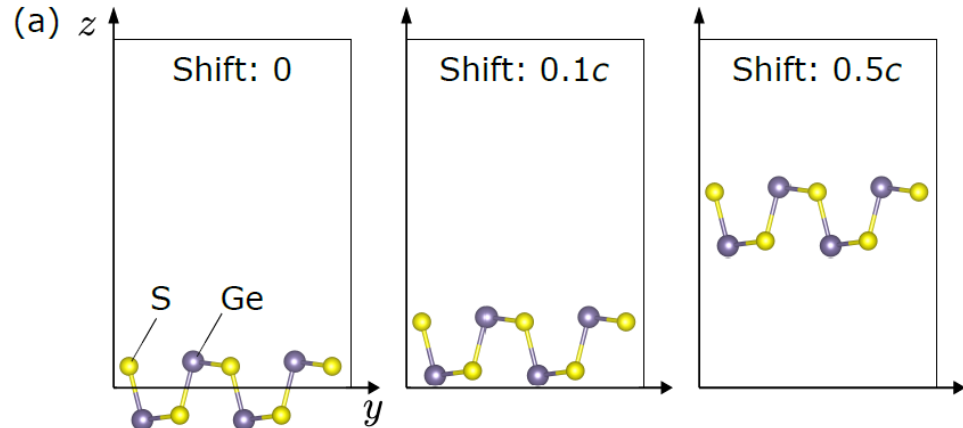
(d) Off-diagonal elements



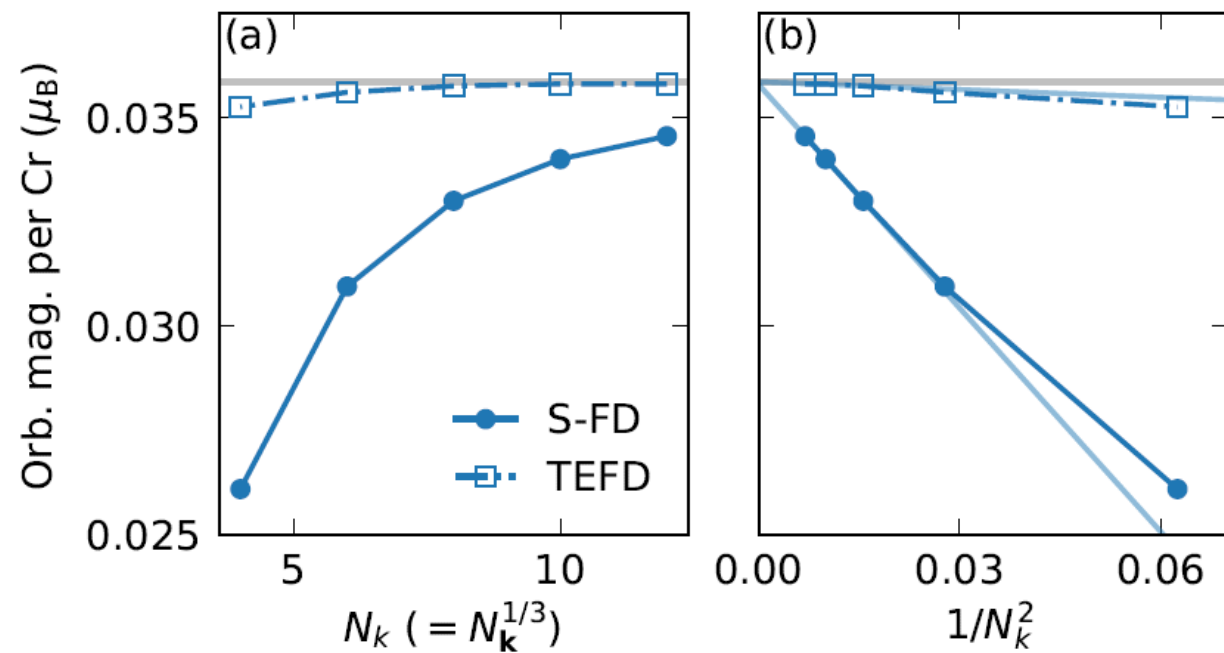
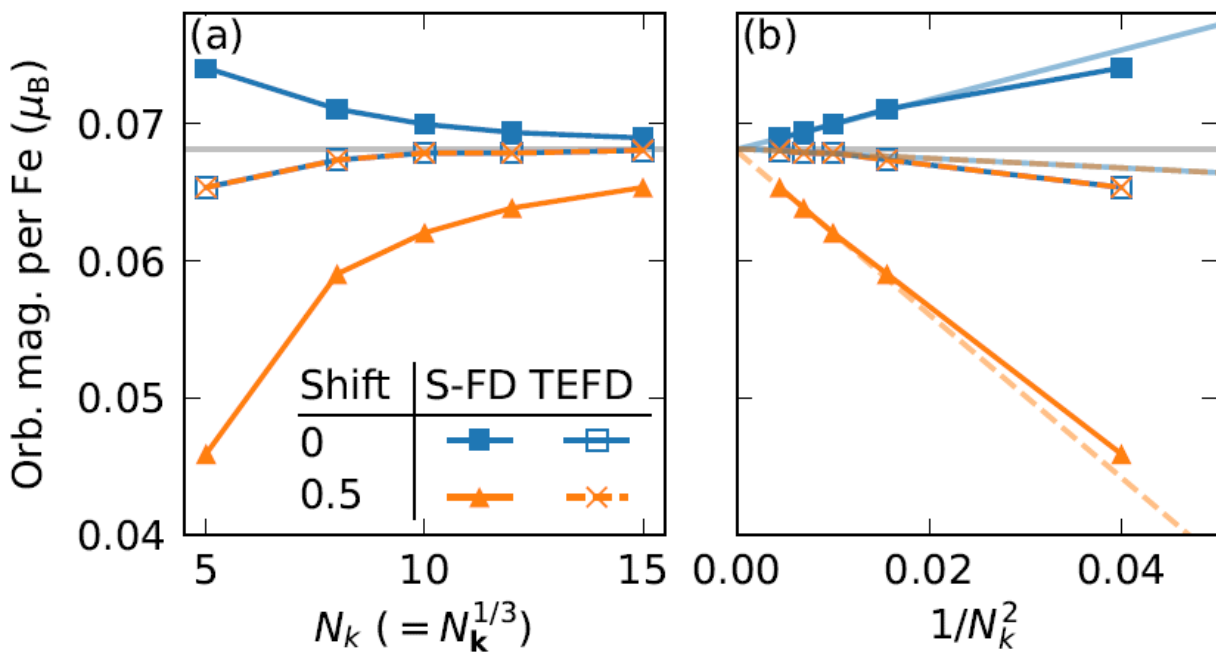
..... First-order finite difference

- · - Second-order finite difference

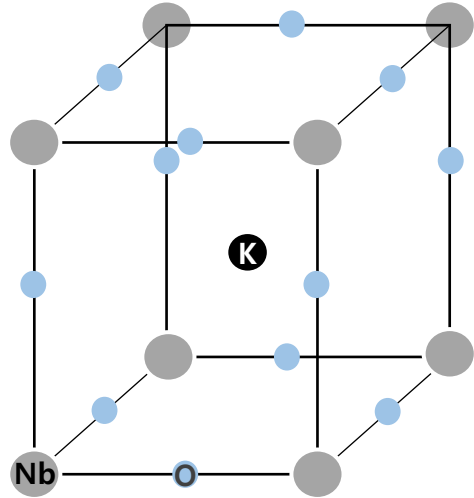
— Third-order finite difference



Orbital magnetization: bcc iron & CrO₂



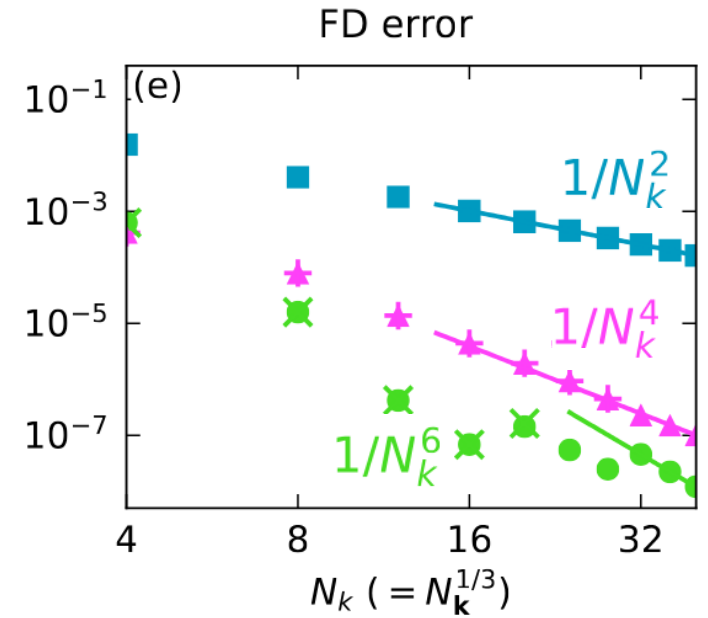
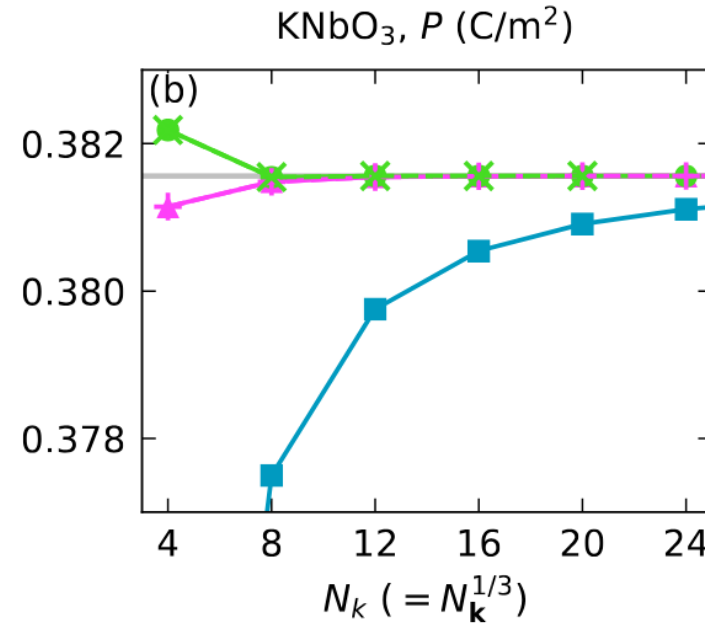
Polarization of KNbO₃



$$\mathbf{P} = \frac{e}{V} \left(\sum_I Z_I \mathbf{r}_I - 2 \sum_{n \in \text{occ.}} \mathbf{r}_n \right)$$

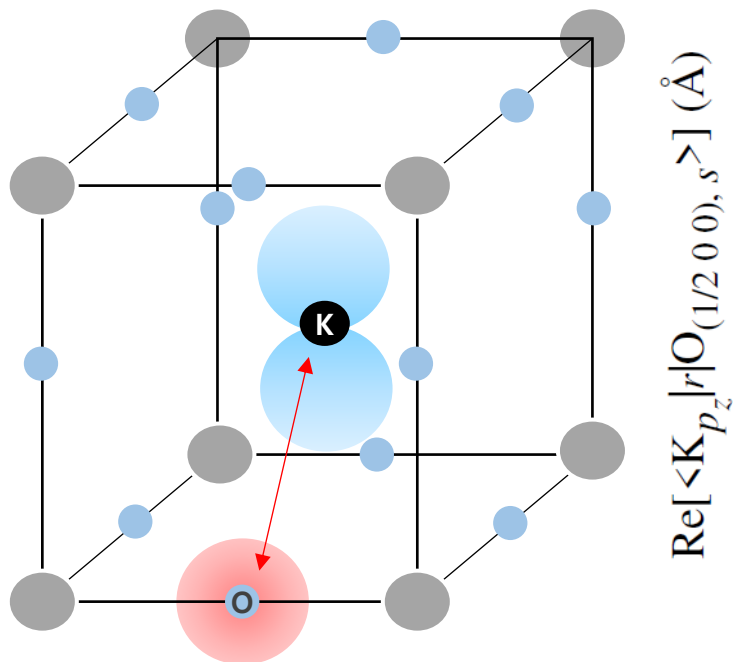
We expect:

$$\begin{aligned} P_z &= \text{const.} + O(b^{2N}) \\ &= \text{const.} + O(N_k^{-2N}) \end{aligned}$$

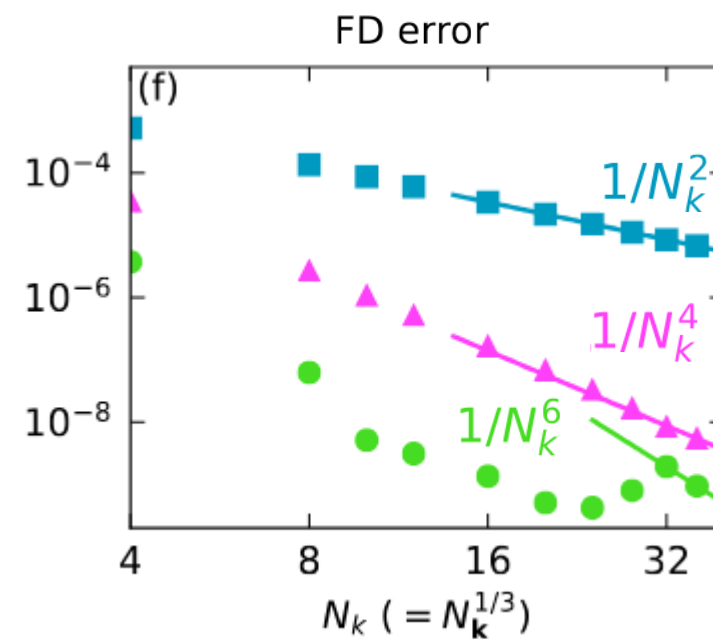
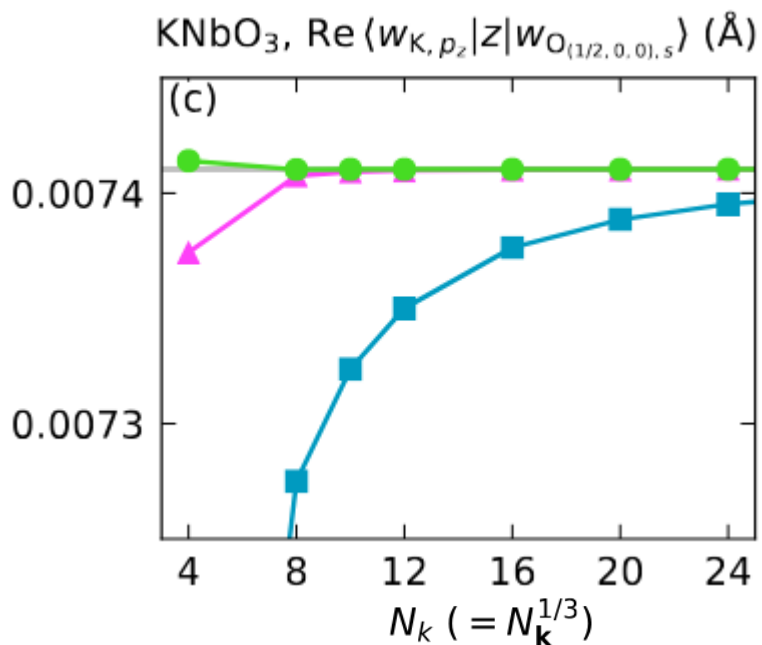


Order	Multiples Shells	
1		
2		
3		

Off-diagonal position components for KNbO_3



$$\text{Re}[\langle K_{p_z} | r | O_{(1/2, 0, 0), s} \rangle] (\text{\AA})$$



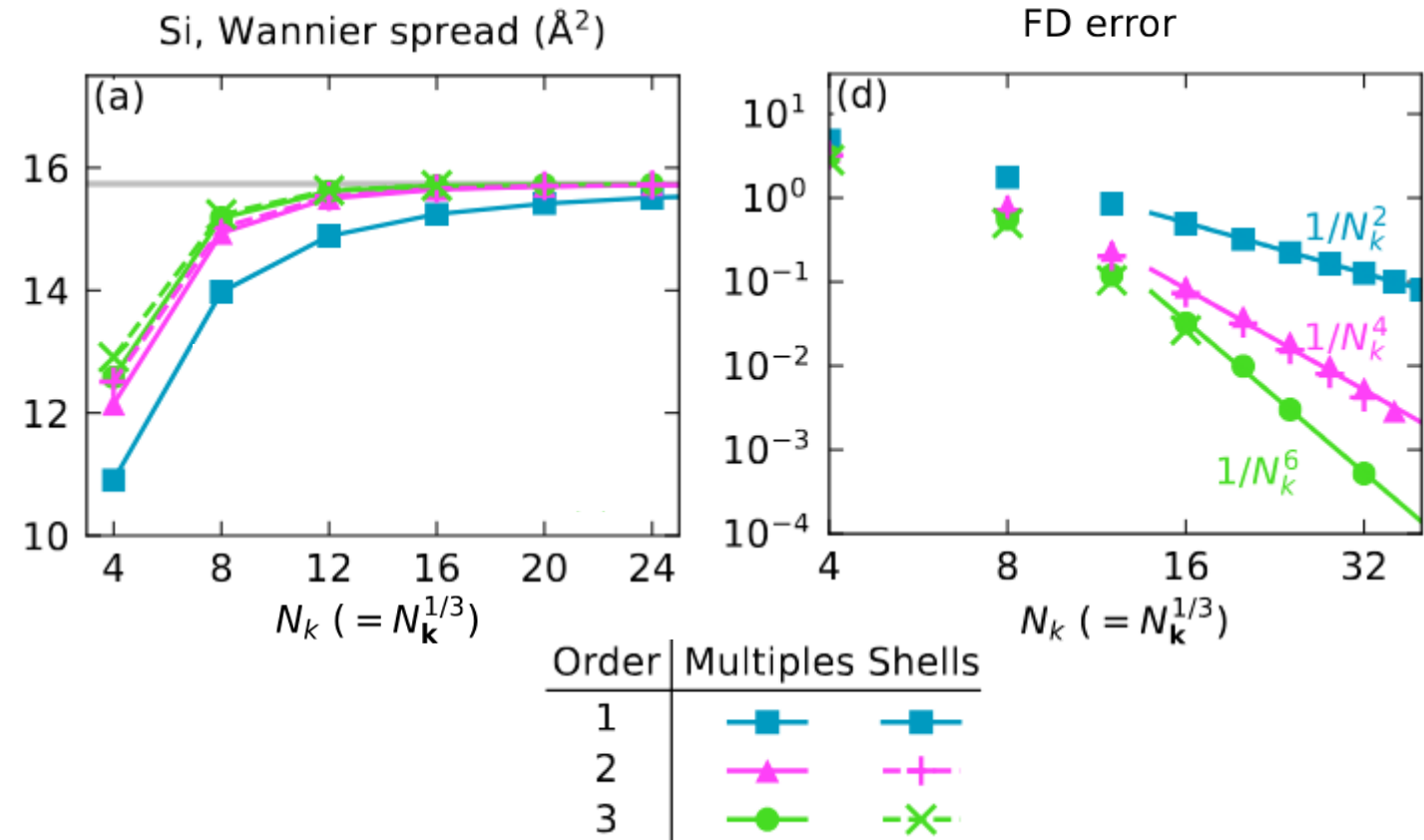
Order	Multiples
1	—■—
2	—▲—
3	—●—

Wannier spread of silicon

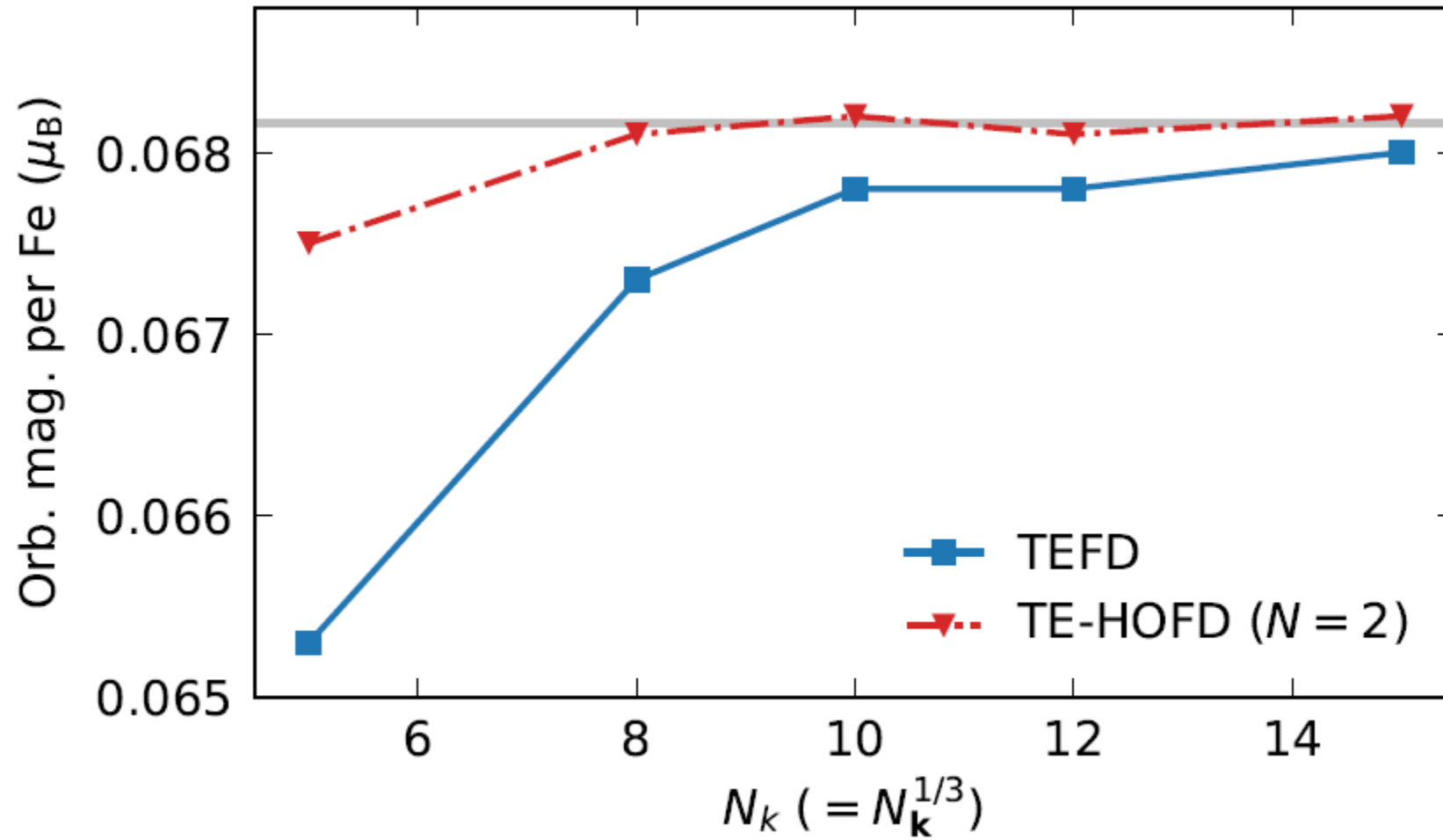
We expect:

$$\Omega = \sum_n [\langle r_n^2 \rangle - \langle \mathbf{r}_n \rangle^2]$$

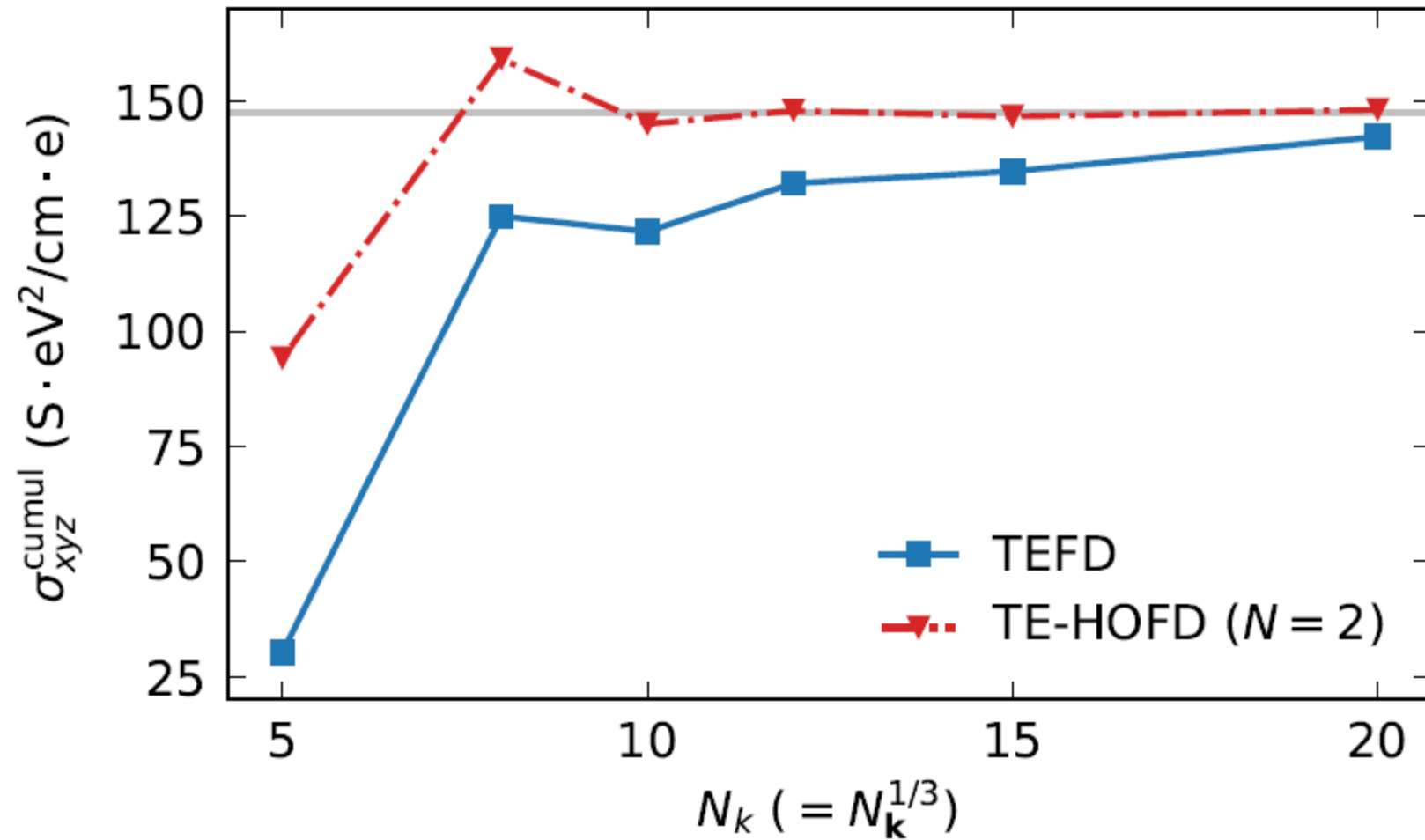
$$= \text{const.} + O(b^{2N})$$



Orbital magnetization of bcc iron

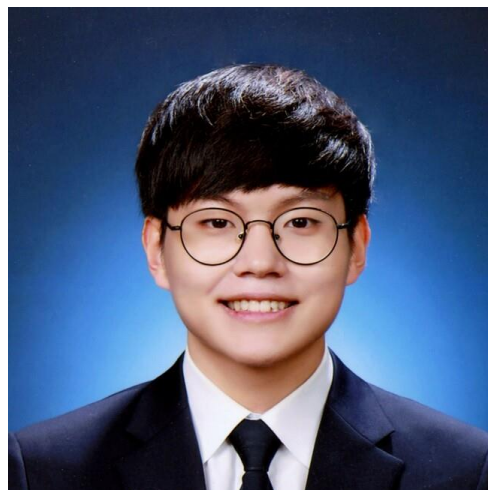


Spin-Hall conductivity of GaAs





Jae-Mo Lihm
SNU (now @ Flatiron)



Minsu Ghim
SNU



Seung-Ju Hong
SNU

Conclusion

Relevant for operators involving the momentum derivative of u .

Translationally-equivariant finite difference (TEFM) method significantly reduces computational cost **always (even for the orbital magnetization of bcc iron, whose single atom is at the origin)**

Even more so if the Wannier functions are far from the origin, or if the unit cell is large (nanostructures, defect, ...)

2nd order finite difference method is powerful when combined with TEFD formalism

These features are implemented in

WANNIER90

WANNIER BERRI

J.-M. Lihm, M. Ghim, S.-J. Hong, and C.-H. Park, arXiv:2604.22614